Koneru Lakshmaiah Education Foundation
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Department of CSE

B.Tech. CSE Trimester -5

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Cloud Infrastructure and Services

25CS2206E

LAB RECORD

2026

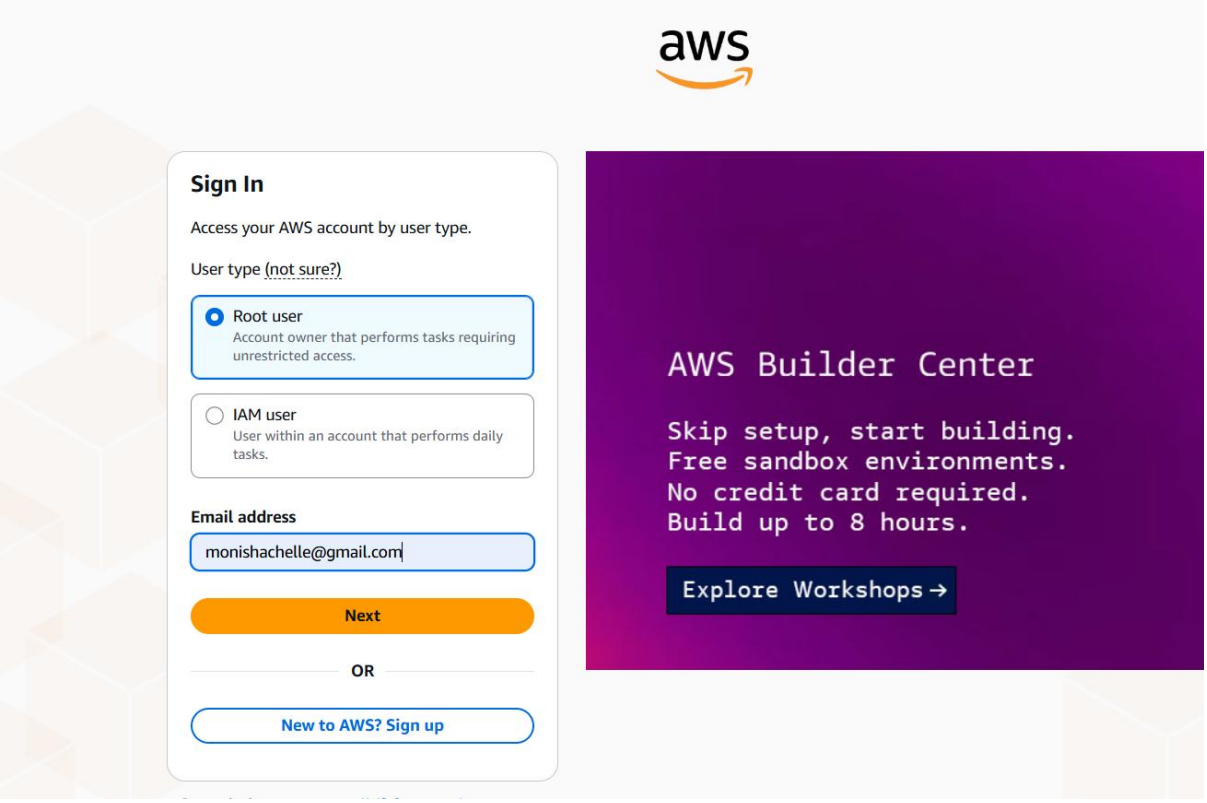
Experiment -2

AIM:

2 a) Enabling Multi-Factor Authentication to Secure Your Access
b) Creating First Linux Instance
c) S3 Bucket Creation and Upload Objects
I. Create an S3 Bucket for file storage.
II. Upload 5 objects with different file extensions.

Description:

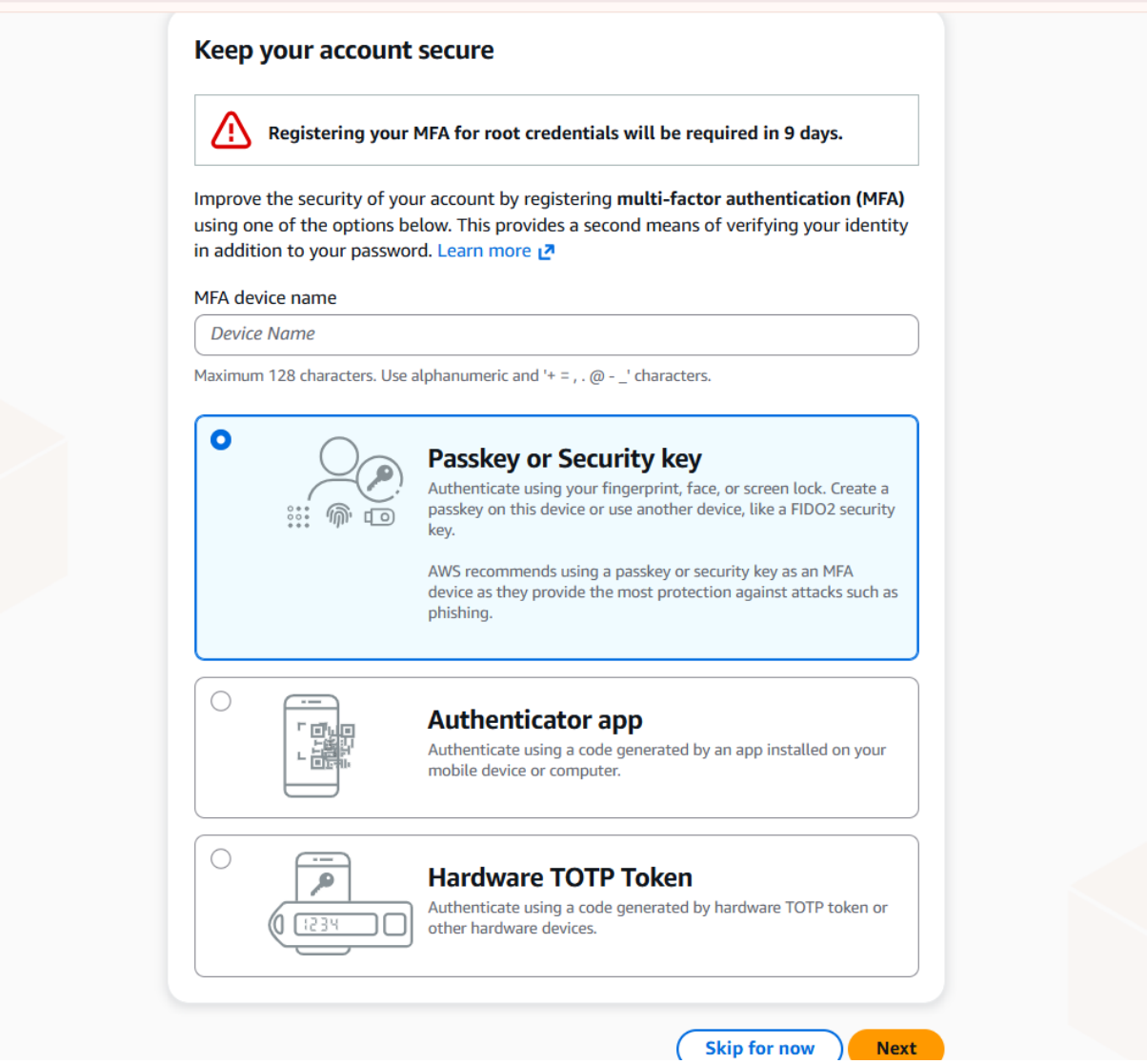
2A. Creating and Configuring the First Linux EC2 Instance (My-First-EC2)



AWS Root User Sign-In

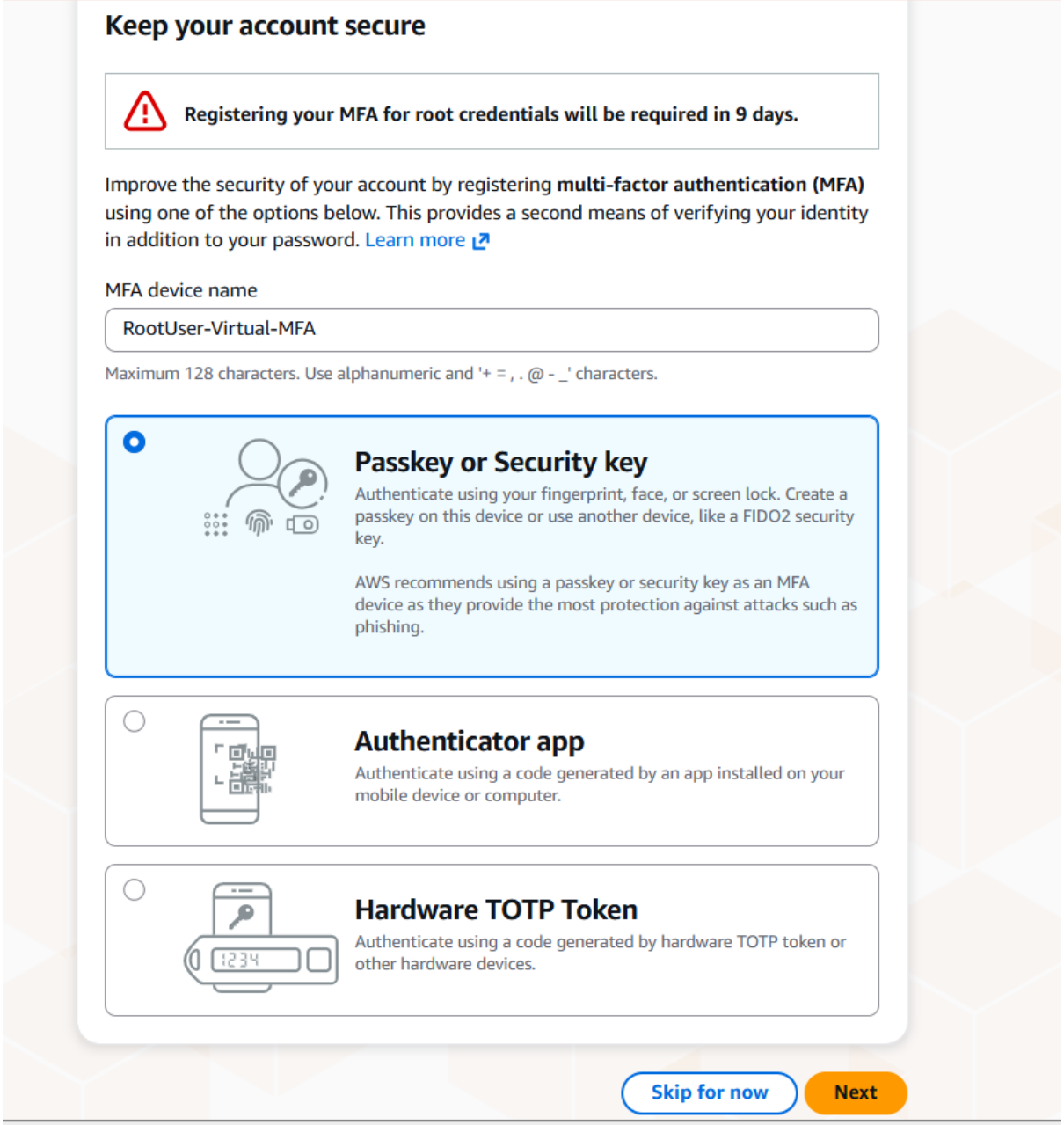
Description: *Opened the AWS Management Console sign-in page and selected the Root user option. Entered the registered AWS account email address to begin signing in and access the AWS services.*

Result: *The AWS Root user sign-in process was started successfully*



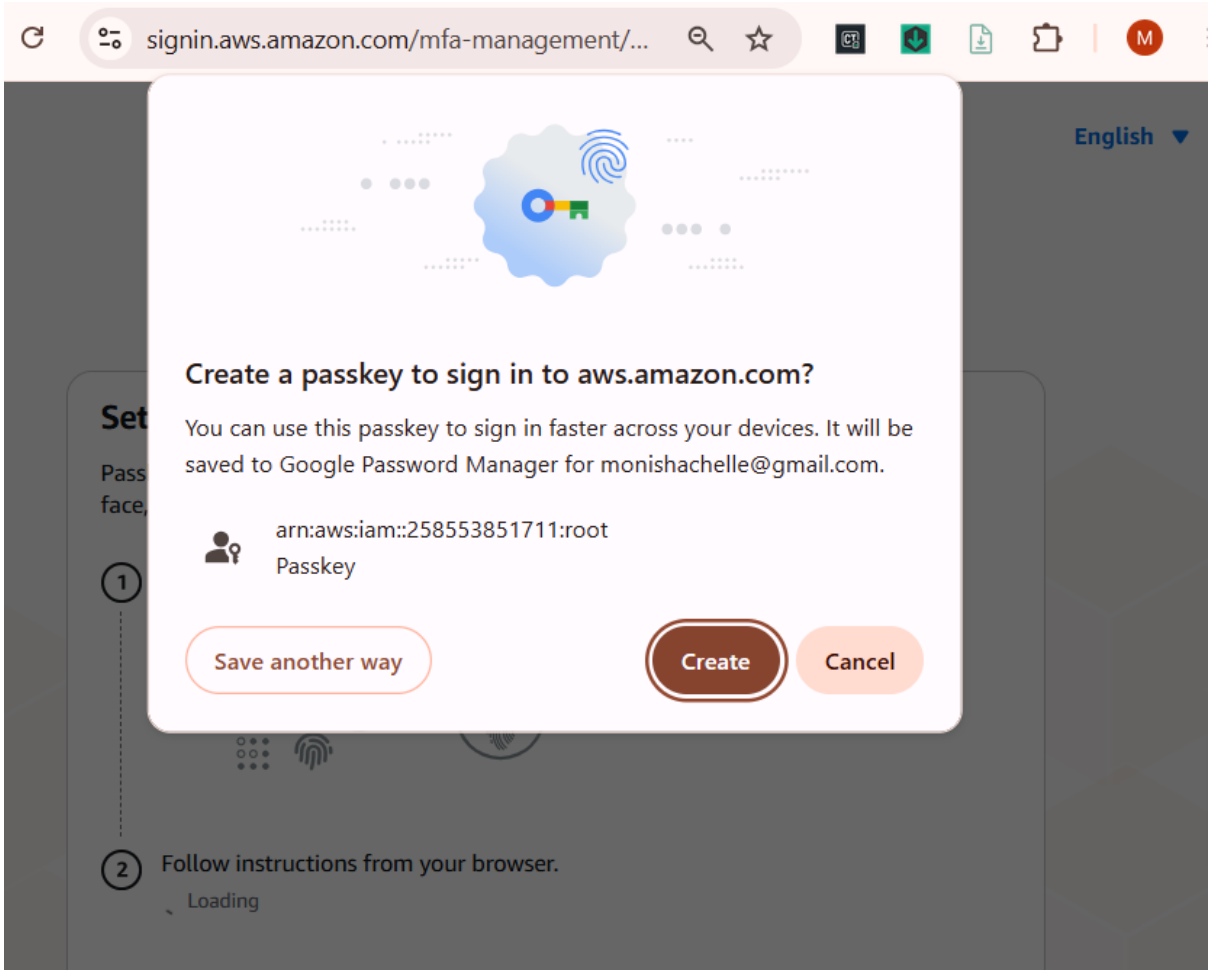
AWS MFA setup page

Description: *After signing in to the AWS Root account, the account security page displayed the requirement to register Multi-Factor Authentication (MFA). It provided different MFA options, including a passkey, authenticator app, and hardware TOTP token.*



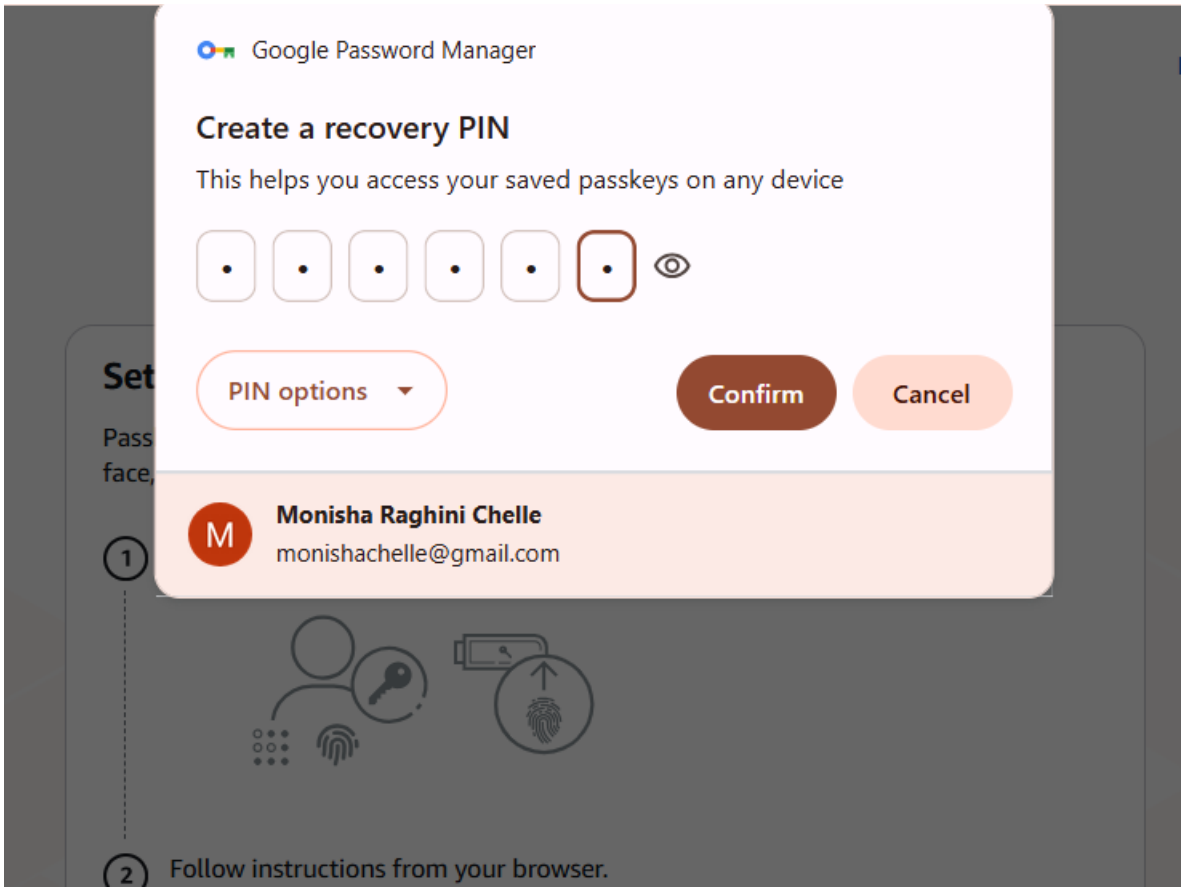
Selecting the MFA device

Description: *Entered the MFA device name as RootUser-Virtual-MFA and selected the Passkey or Security key option to configure an additional authentication method for the AWS Root account.*



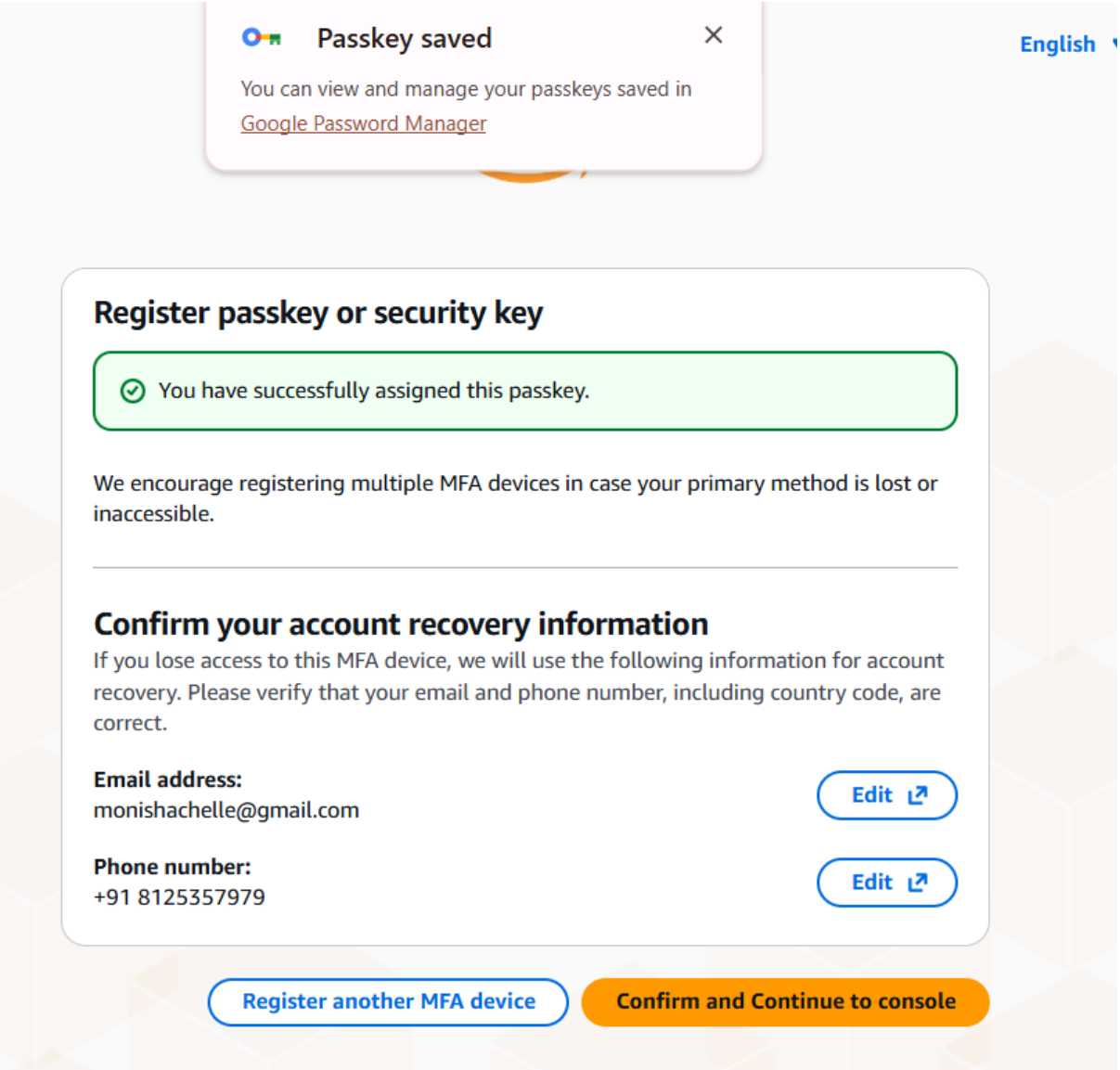
Creating a passkey

Description: *AWS prompted me to create a passkey using Google Password Manager. Selected the Create option to save the passkey for secure authentication.*



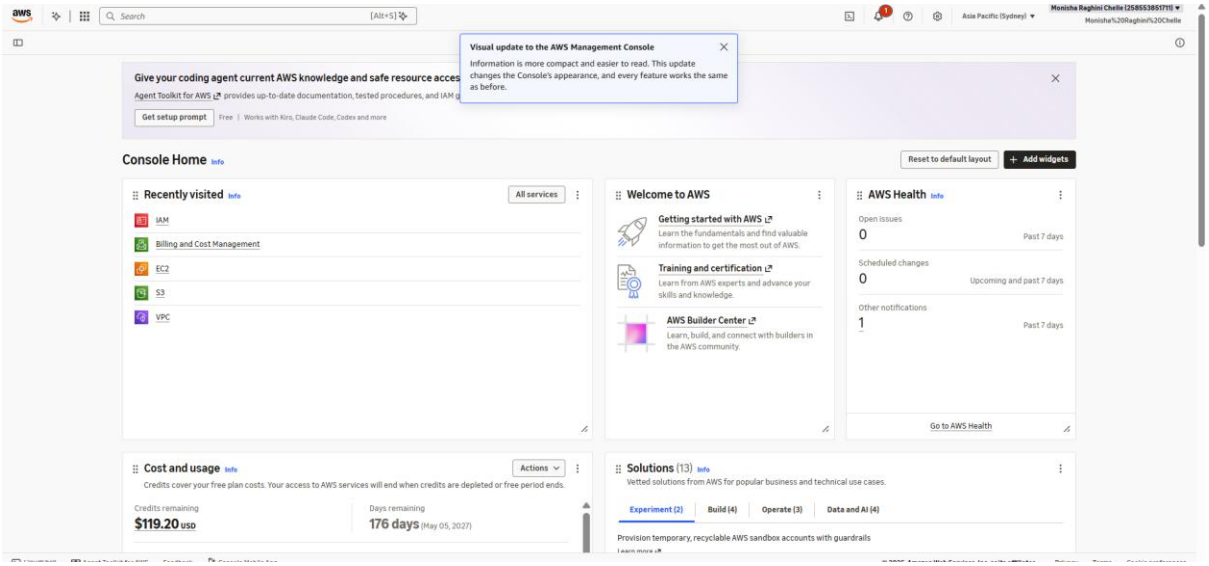
Setting a recovery PIN

Description: *Google Password Manager requested a recovery PIN to help protect and access the saved passkey. Entered the PIN and confirmed it to continue the setup.*



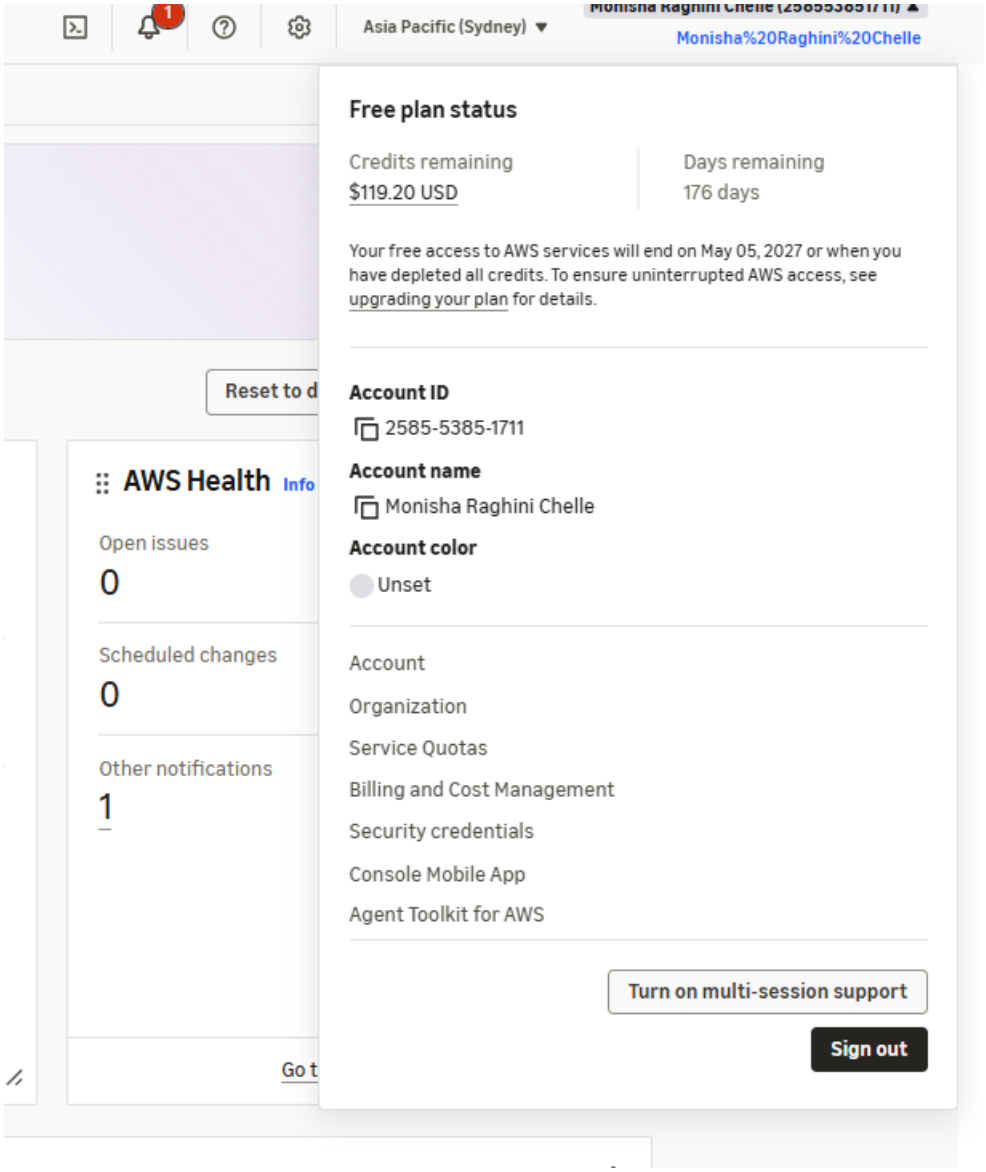
Successful MFA registration

Description: *AWS displayed a confirmation message stating that the passkey was successfully assigned. The account recovery information was also displayed for verification.*



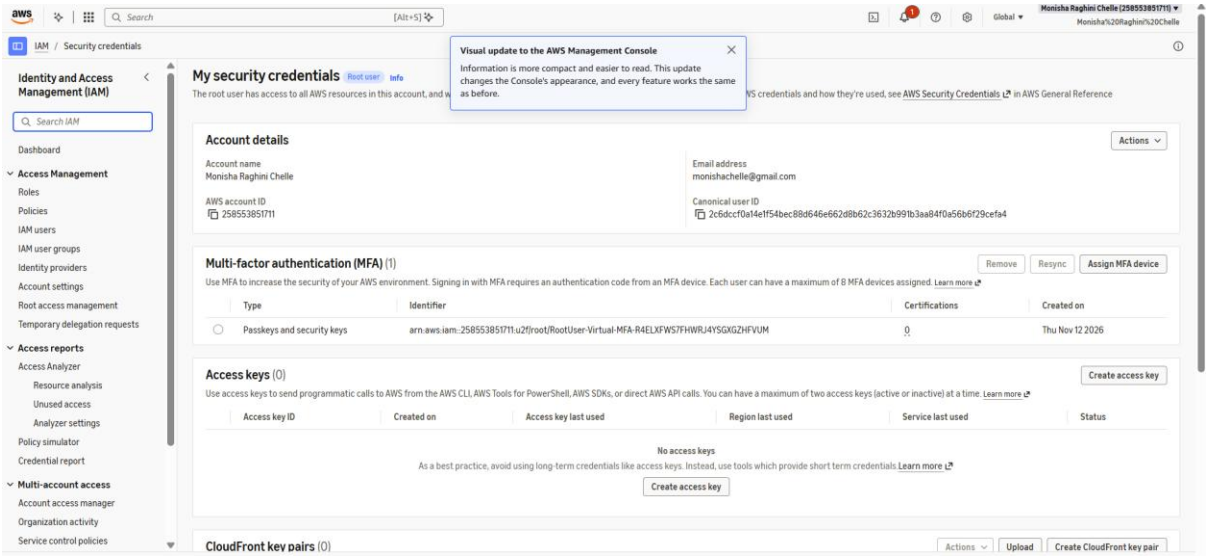
AWS Management Console

Description: *After successfully registering the passkey, selected Confirm and Continue to console to access the AWS Management Console. The console home page was displayed successfully.*



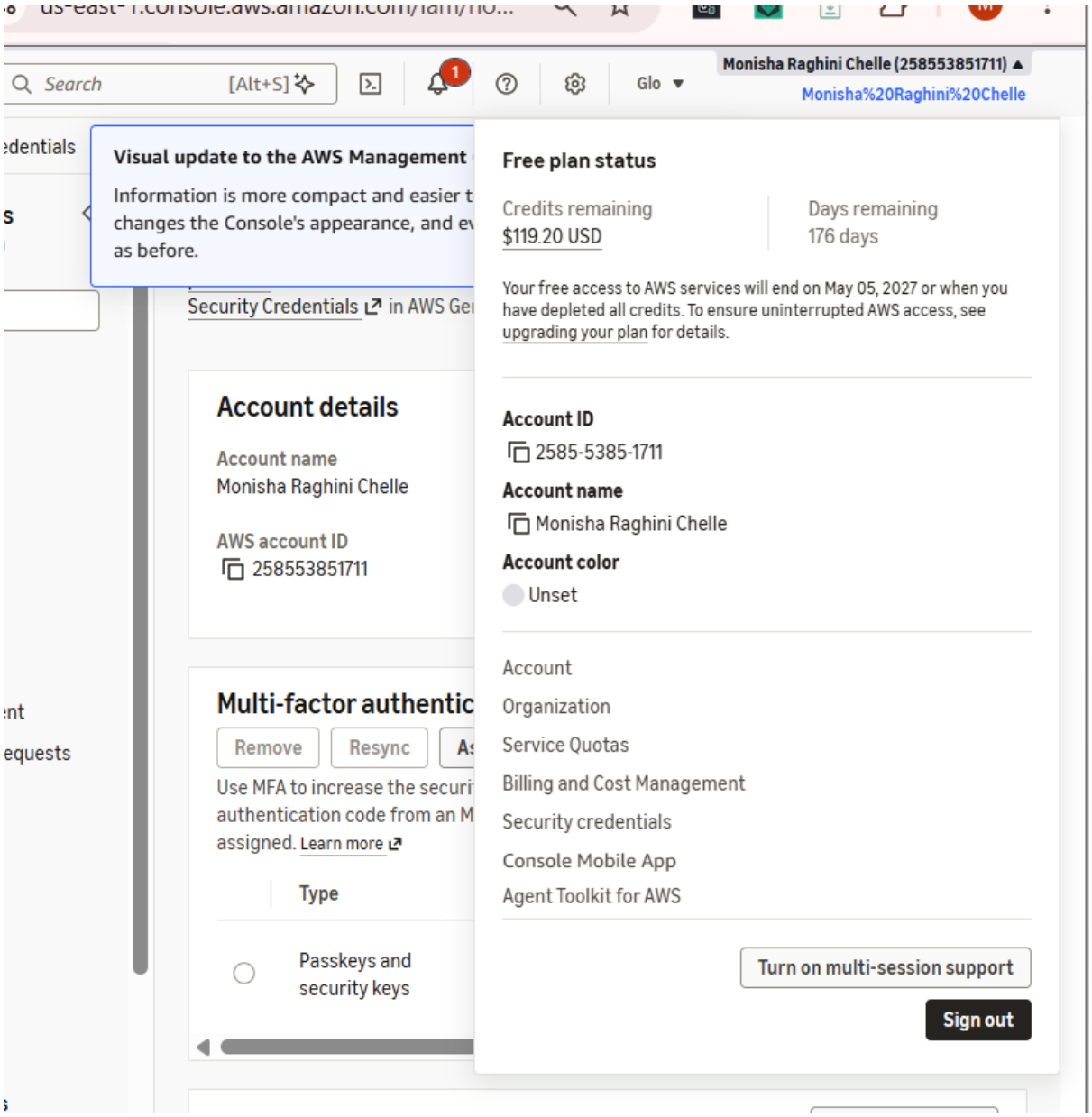
AWS Account Details

Description: *After successfully registering the root user passkey, accessed the AWS Management Console and opened the account menu. Verified the AWS account details, including the account name, remaining AWS credits, and free plan validity period.*



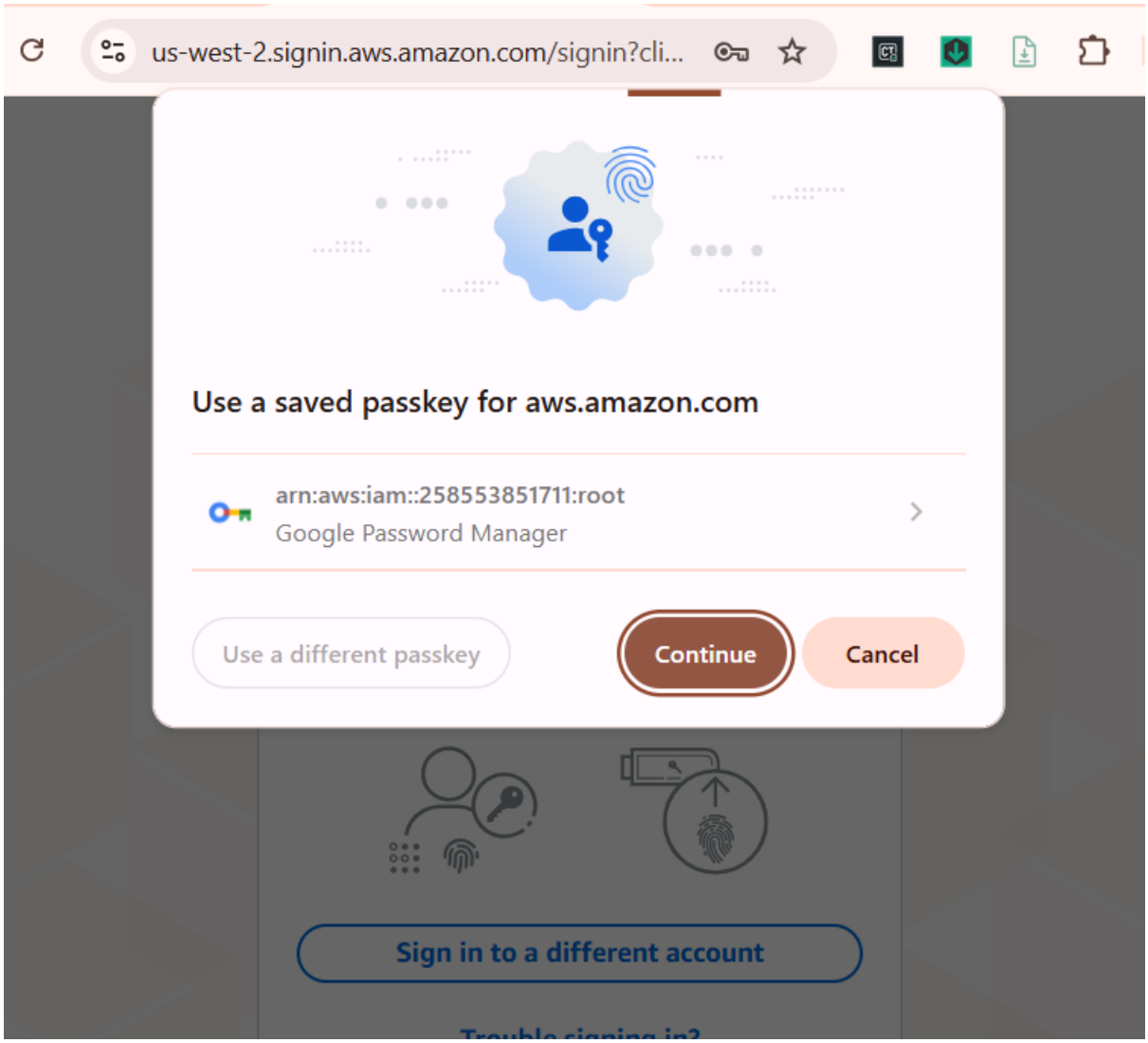
AWS IAM Security Credentials

***Description**: Opened the IAM Security Credentials page to verify the root user's security settings. Confirmed that Multi-Factor Authentication (MFA) was successfully registered using a passkey and checked the access key section, which showed that no access keys were currently configured.*



AWS Account Sign Out

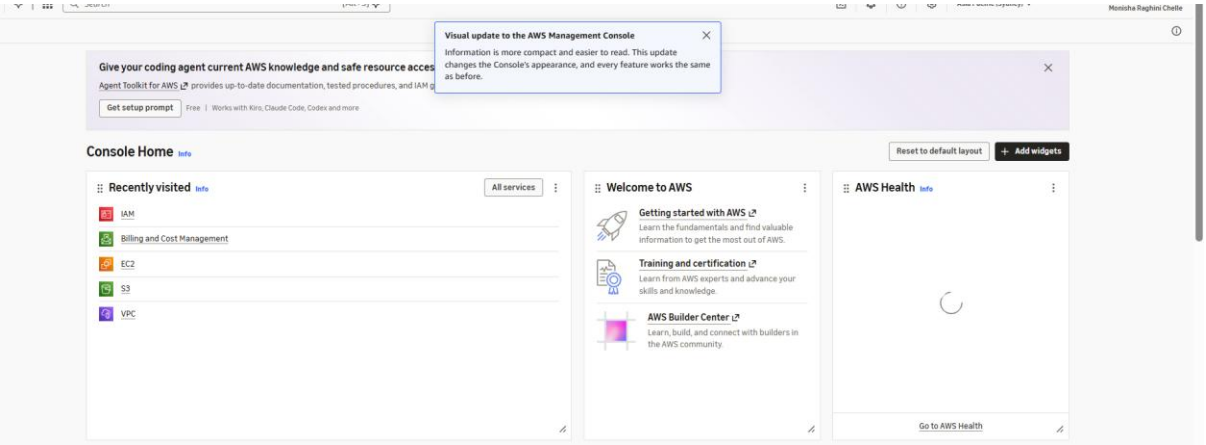
Description: *After completing the AWS account security verification, clicked the Sign out option from the account menu to securely log out of the AWS Management Console.*



AWS Root User Passkey Sign-In

Description: *After signing out of the AWS Management Console, the saved root user passkey appeared through Google Password Manager. Selected the saved passkey and clicked Continue to verify the root user's identity and sign in securely.*

2B. Connecting to the Linux EC2 Instance Using SSH



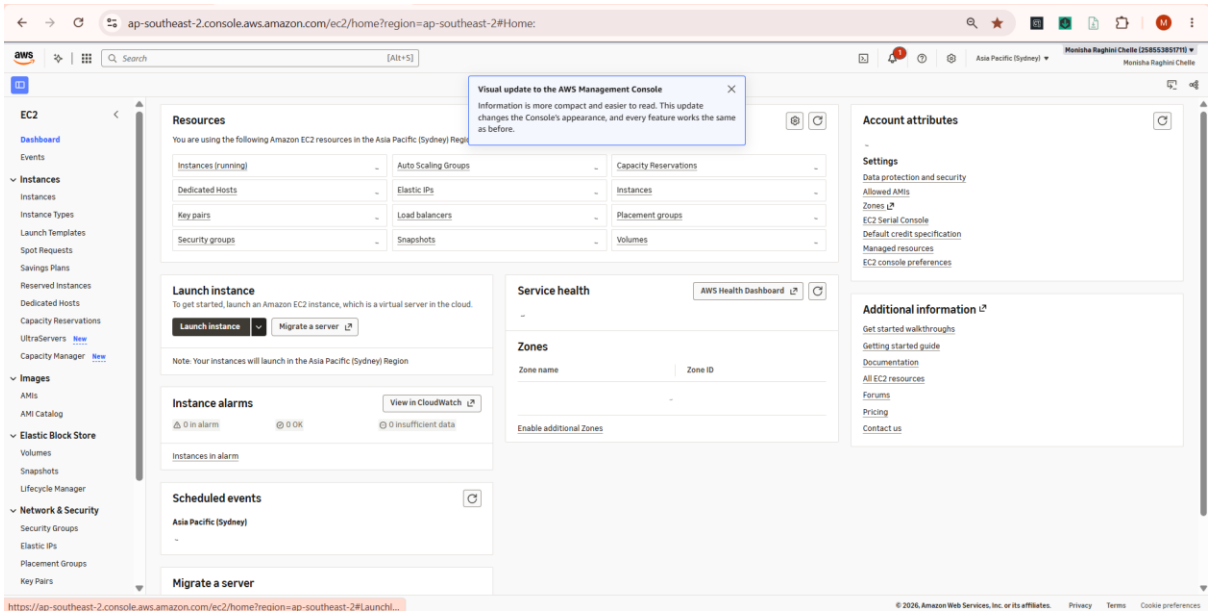
→Click on EC2

AWS Management Console

Description: *After successfully signing in to the AWS Management Console using the registered passkey, accessed the Console Home page to begin creating a Linux EC2 instance.*

Opening EC2

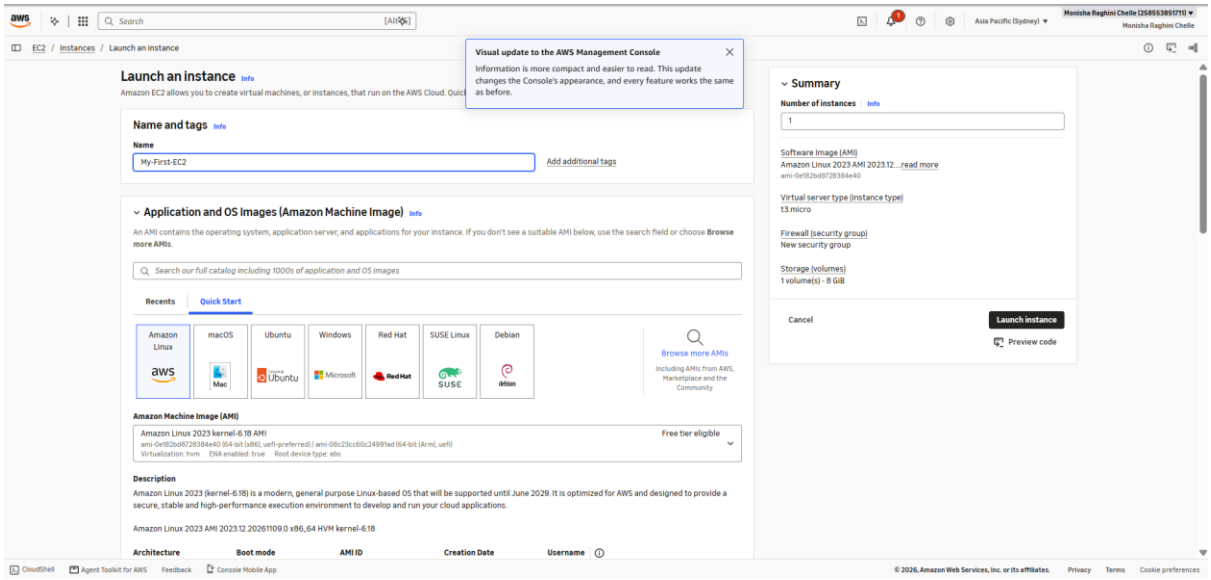
Description: *Selected the EC2 service from the Recently Visited section to access Amazon EC2 and manage virtual server instances.*



→Click on Launch instance

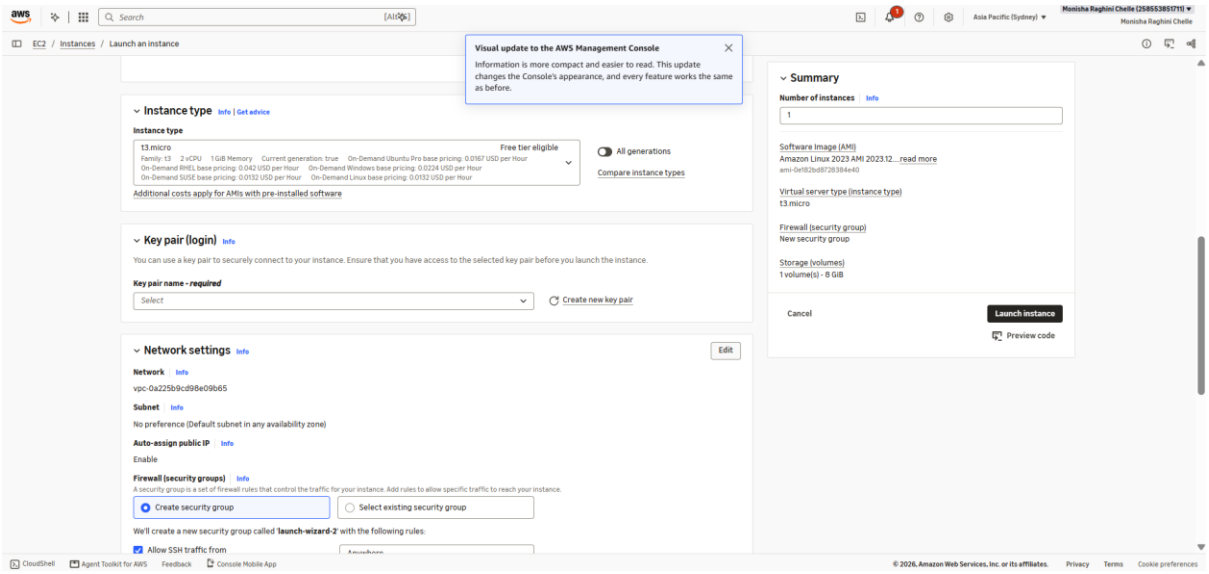
Launching an Instance

Description: *Opened the EC2 Dashboard and selected Launch instance to begin creating a new Linux virtual machine.*



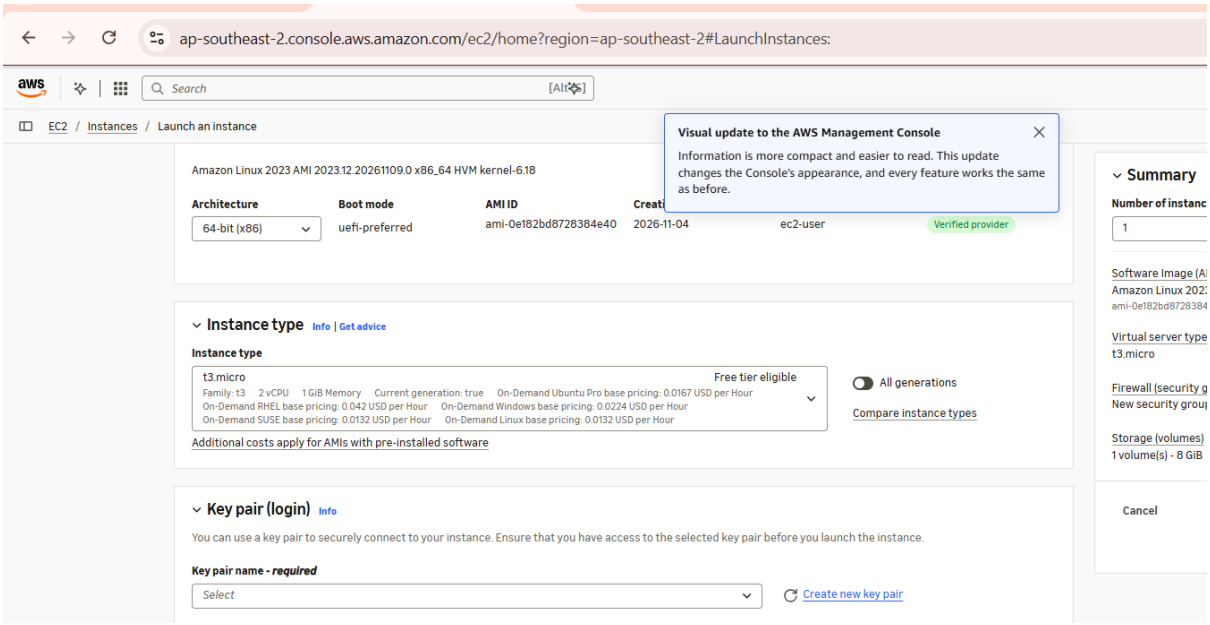
Selecting the Amazon Machine Image (AMI)

Description: *Entered the instance name as My-First-EC2 and selected Amazon Linux as the operating system from the available AMI options.*



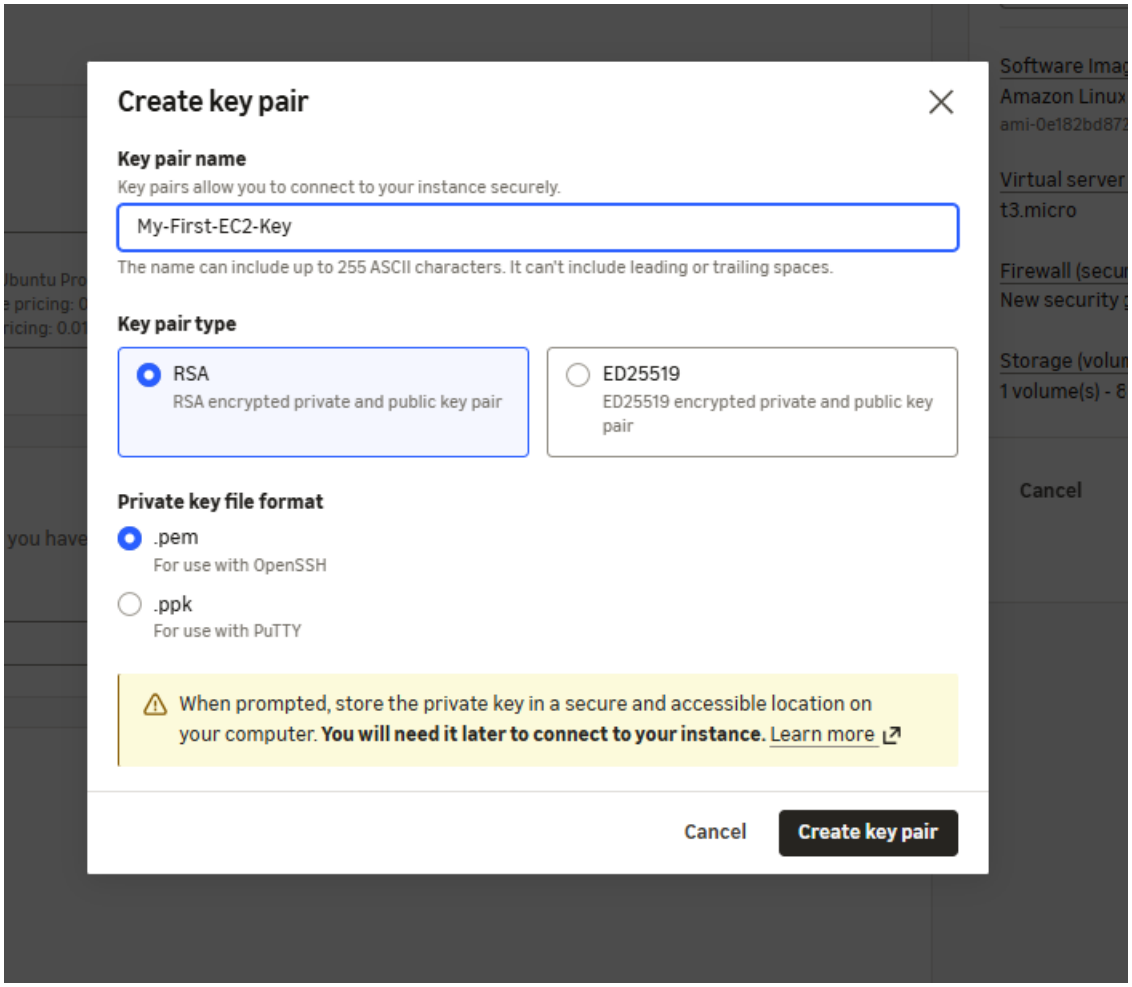
Selecting the Instance Type

Description: *Reviewed the available EC2 instance types and selected t3.micro, which is shown in the instance configuration summary, for the Linux virtual machine.*



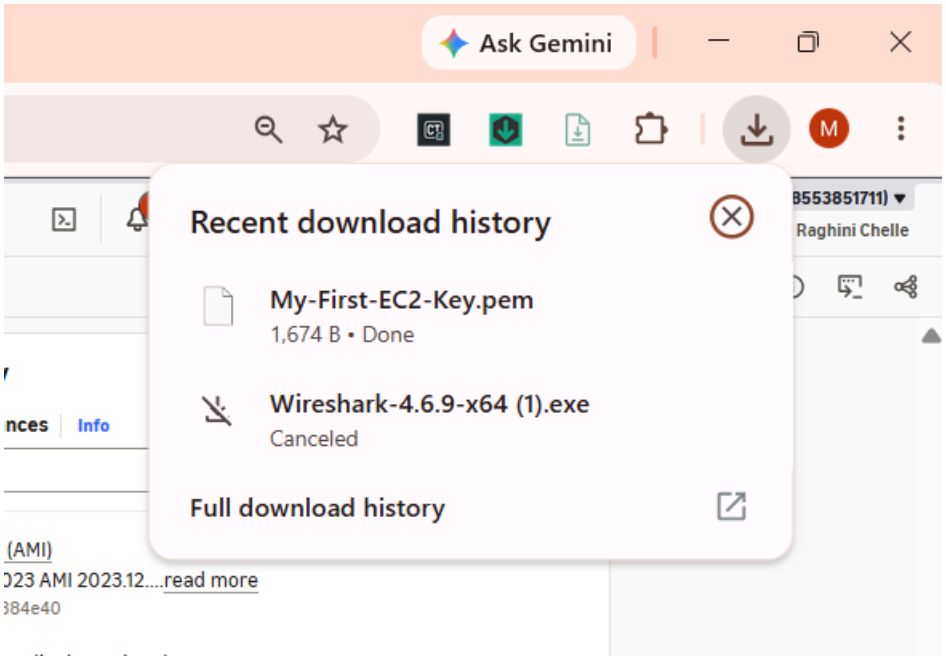
Creating a Key Pair

Description: *Clicked on Create new key pair under the Key pair (login) section to generate a secure key pair for connecting to the EC2 instance.*



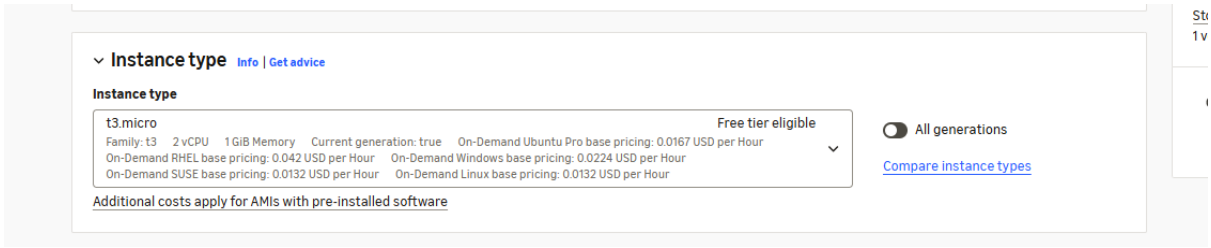
Configuring the Key Pair

Description: *Entered the key pair name as My-First-EC2-Key, selected RSA as the key pair type and .pem as the private key file format.*



Downloading the Key Pair

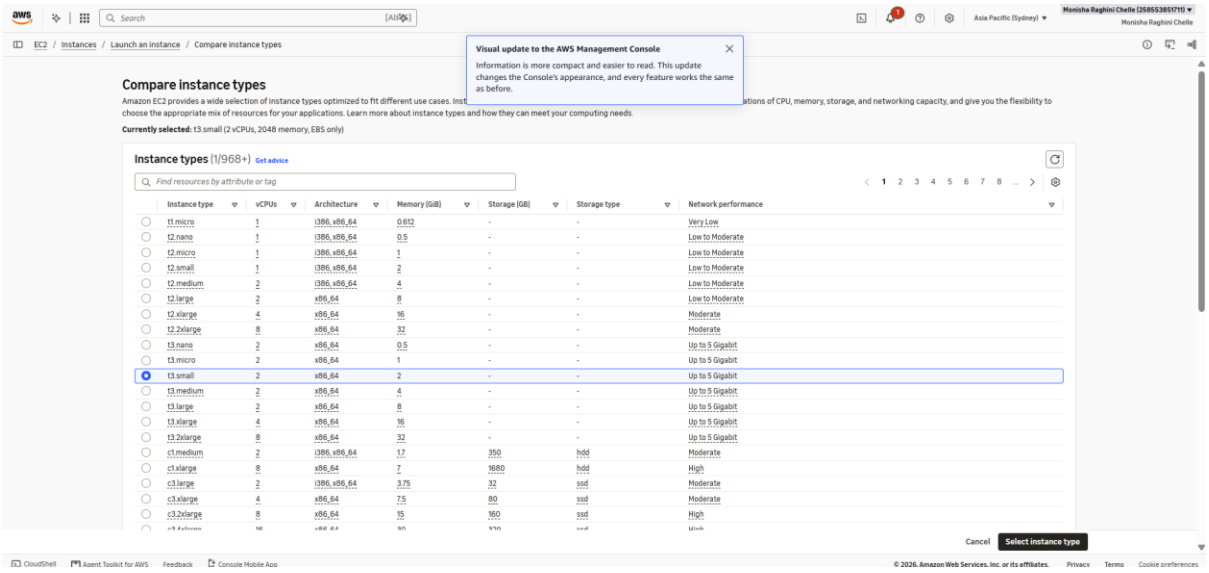
Description: *After creating the key pair, the My-First-EC2-Key.pem file was downloaded successfully. This private key file will be used to securely connect to the EC2 instance.*



Click on Instance type -> Compare instance types

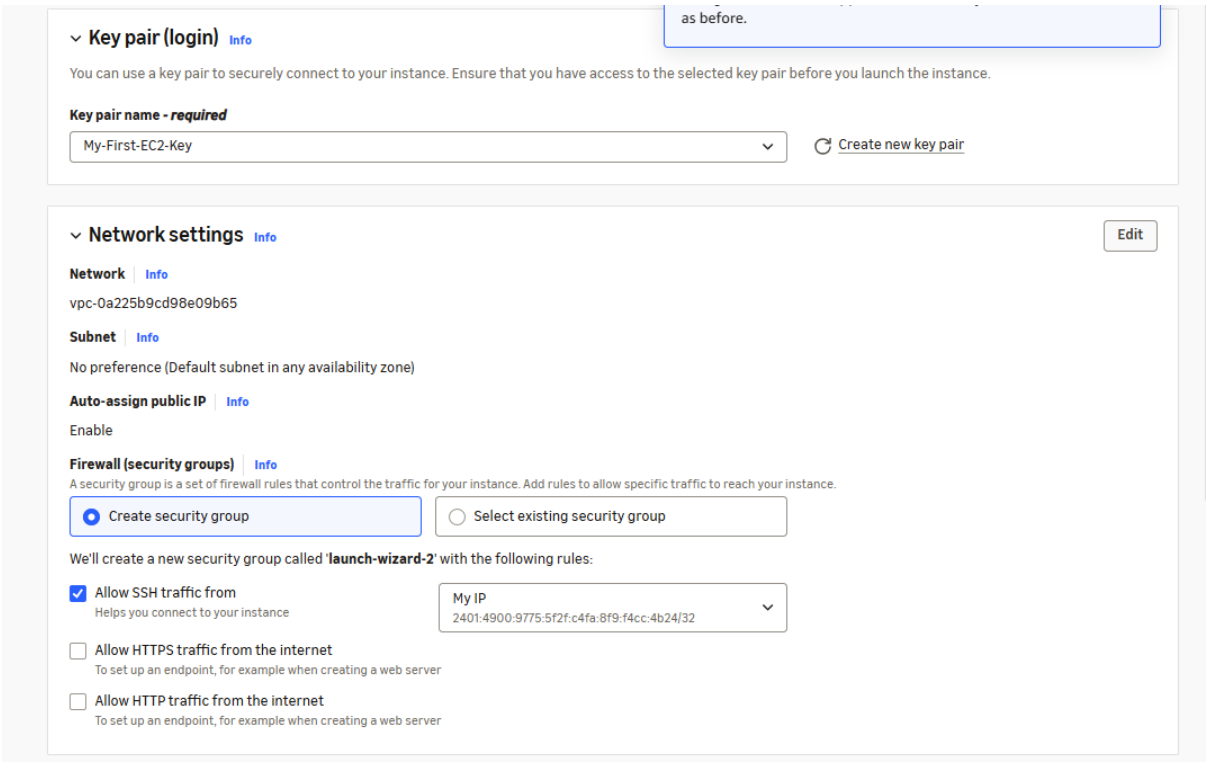
Selecting the EC2 Instance Type

Description: *In the Instance type section, selected the Compare instance types option to view and compare the available EC2 instance types based on their CPU, memory, storage, and network performance.*



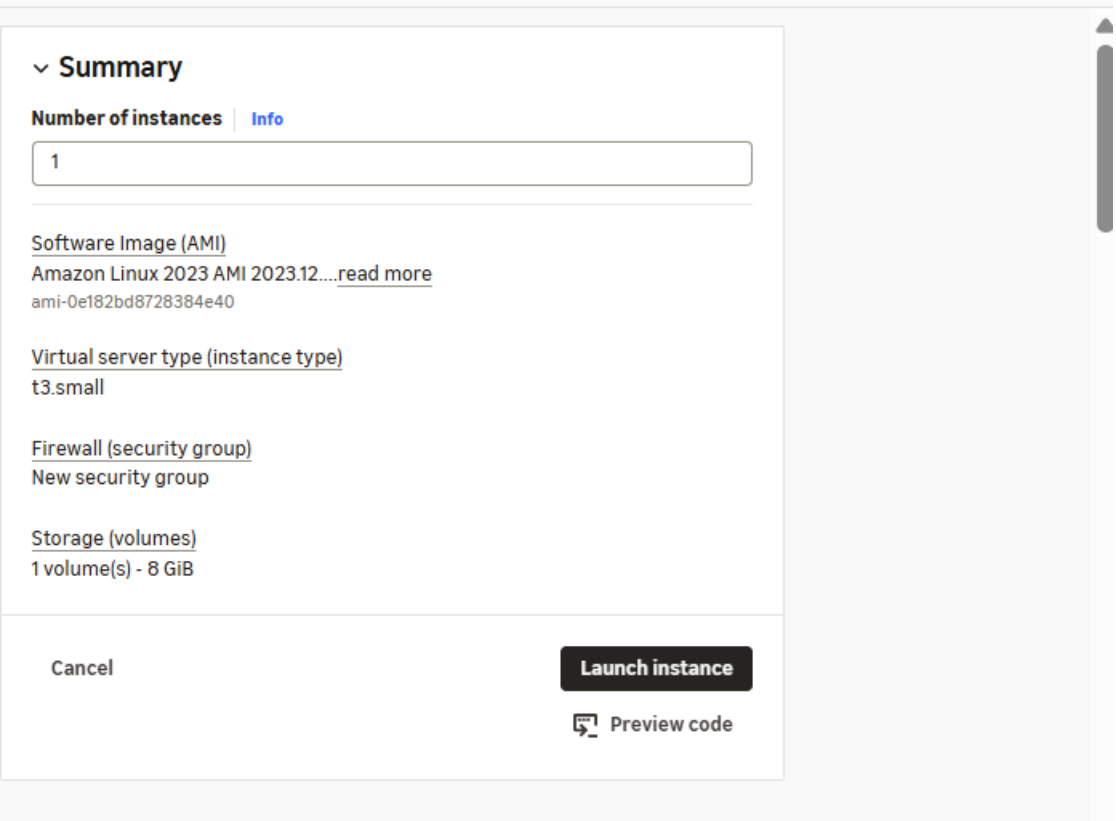
Comparing EC2 Instance Types

Description: *The Compare instance types page displays different EC2 instance types with their specifications, including vCPUs, architecture, memory, storage, and network performance. This helps in selecting a suitable instance type for the Linux virtual machine.*



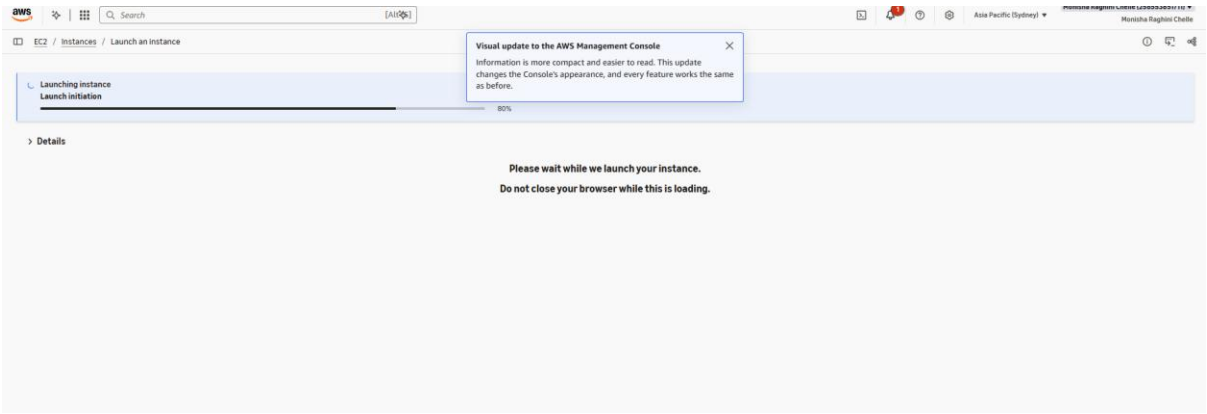
Configuring Network Settings

Description: *Selected the previously created key pair, My-First-EC2-Key, and reviewed the network settings. Kept the default VPC and subnet settings, enabled auto-assign public IP, and selected Create security group with SSH traffic allowed from My IP.*



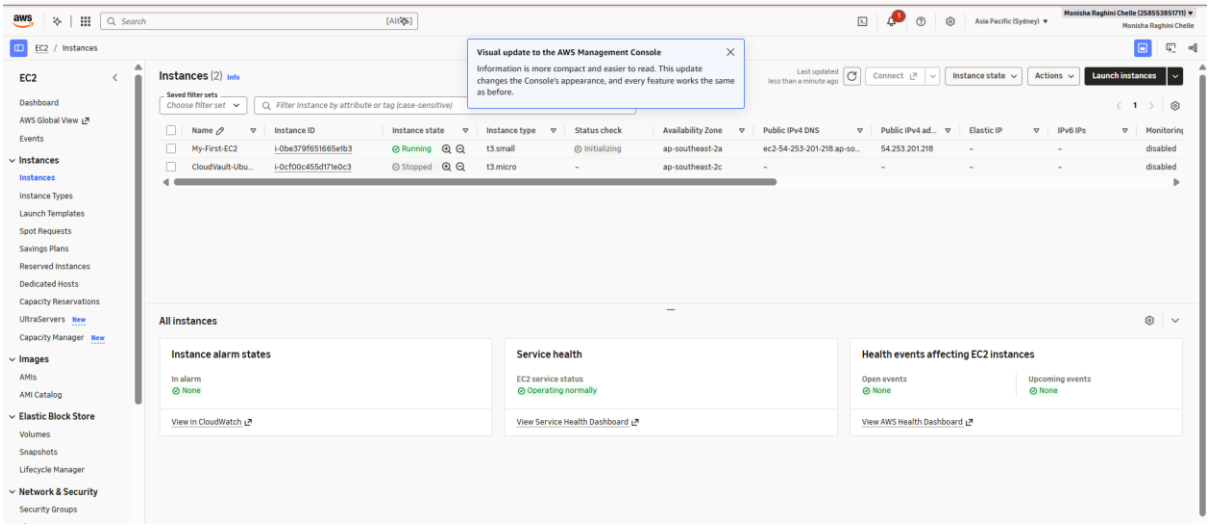
Reviewing Instance Configuration

Description: *Reviewed the EC2 instance launch summary. It displays one Amazon Linux 2023 instance with the t3.small instance type, a new security group, and 8 GiB of storage. Clicked Launch instance to start creating the virtual machine.*



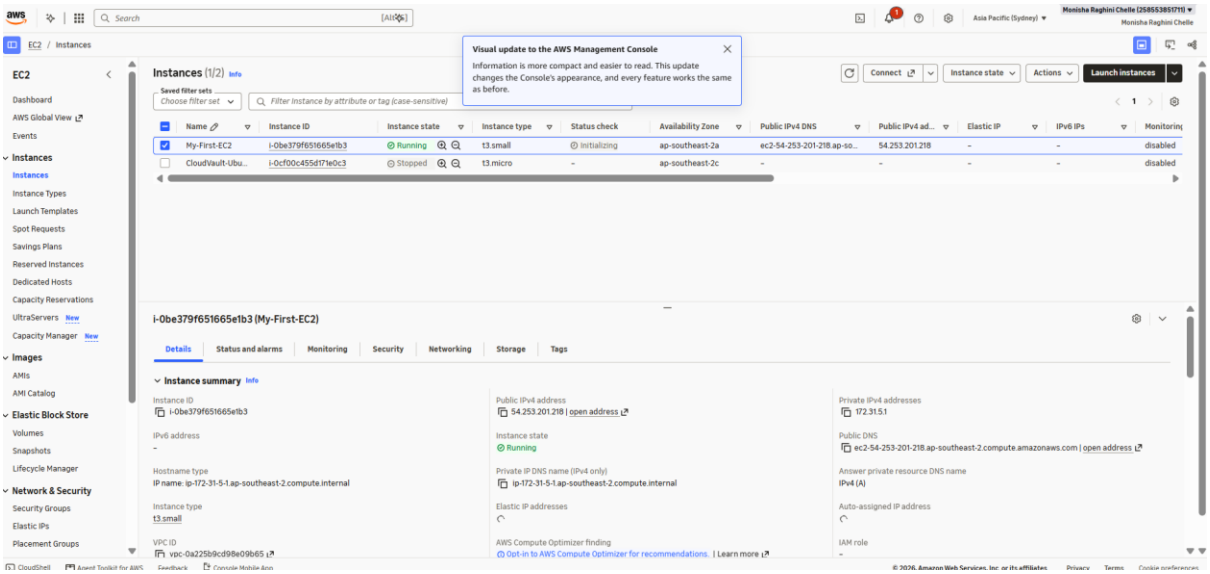
Launching the EC2 Instance

Description: *AWS started the EC2 instance launch process and displayed the launch progress. Waited while AWS initialized the virtual machine.*



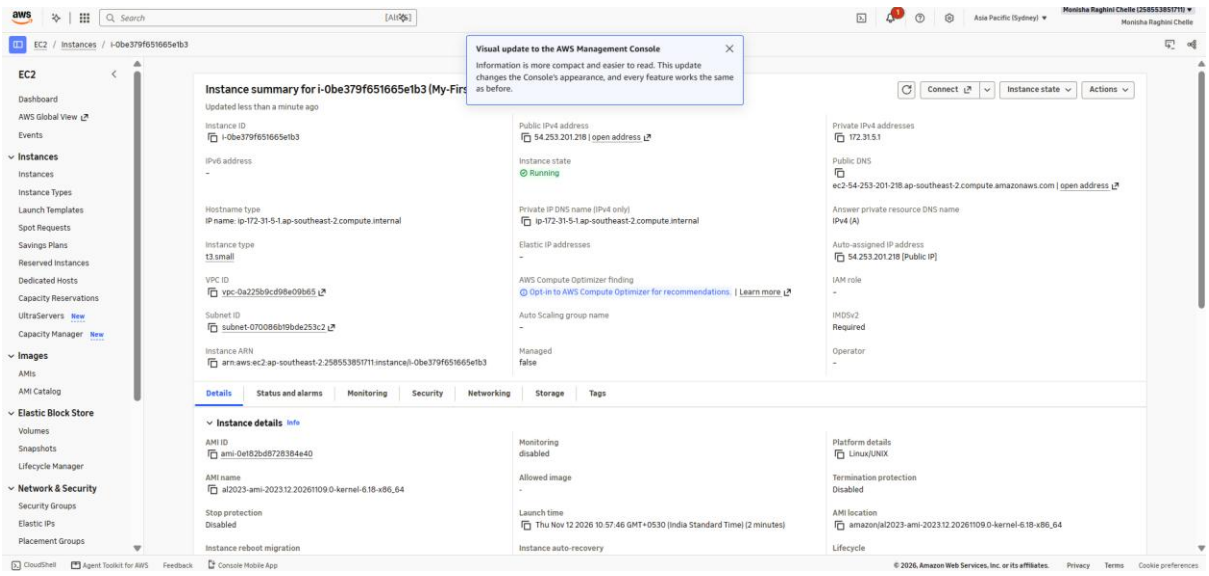
EC2 Instance Launched Successfully

Description: *The EC2 Instances page displays the newly created instance My-First-EC2 with its state shown as Running. This confirms that the Linux virtual machine was launched successfully.*



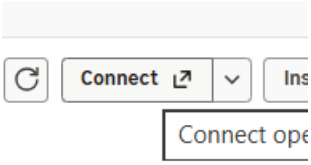
Selecting the EC2 Instance

Description: *Selected the My-First-EC2 instance from the instances list to view its details, including the instance state, public IP address, private IP address, and instance type.*



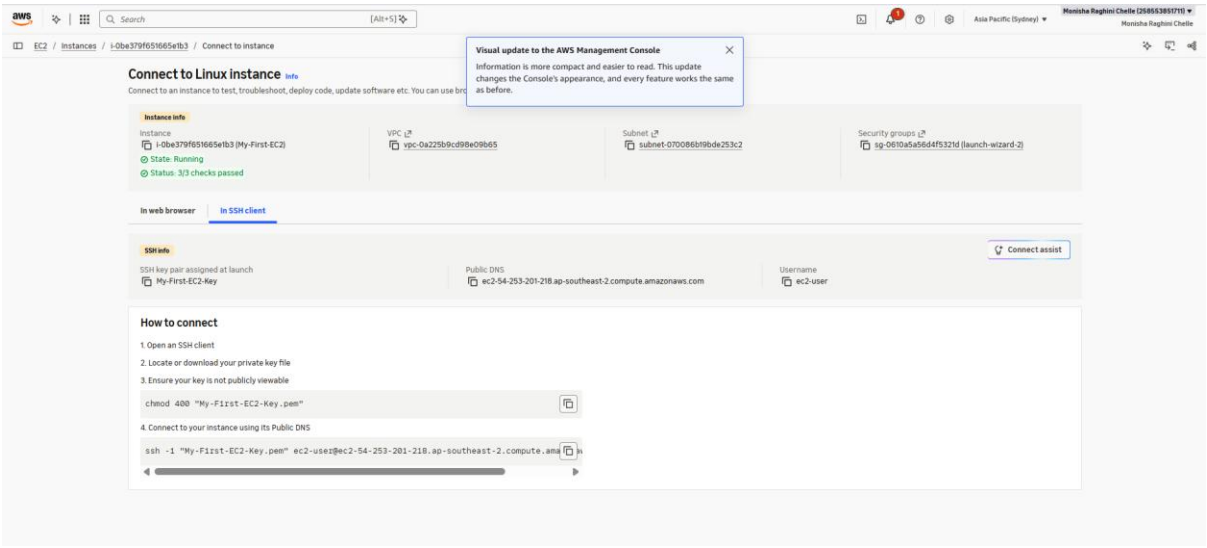
Viewing EC2 Instance Details

Description: *Opened the instance summary page to verify the configuration details of the running EC2 instance, including its Amazon Linux operating system, instance type, IP addresses, and network information.*



Connecting to the EC2 Instance

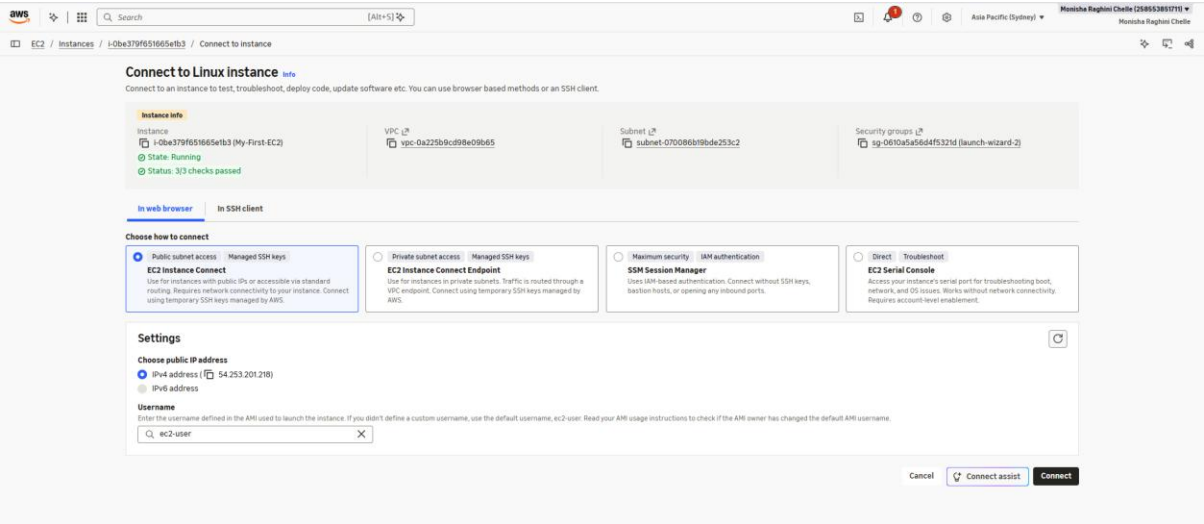
Description: *Located the Connect option on the EC2 instance page. This option is used to access the running Linux instance and begin remote connection.*



Opening the EC2 Connection Page

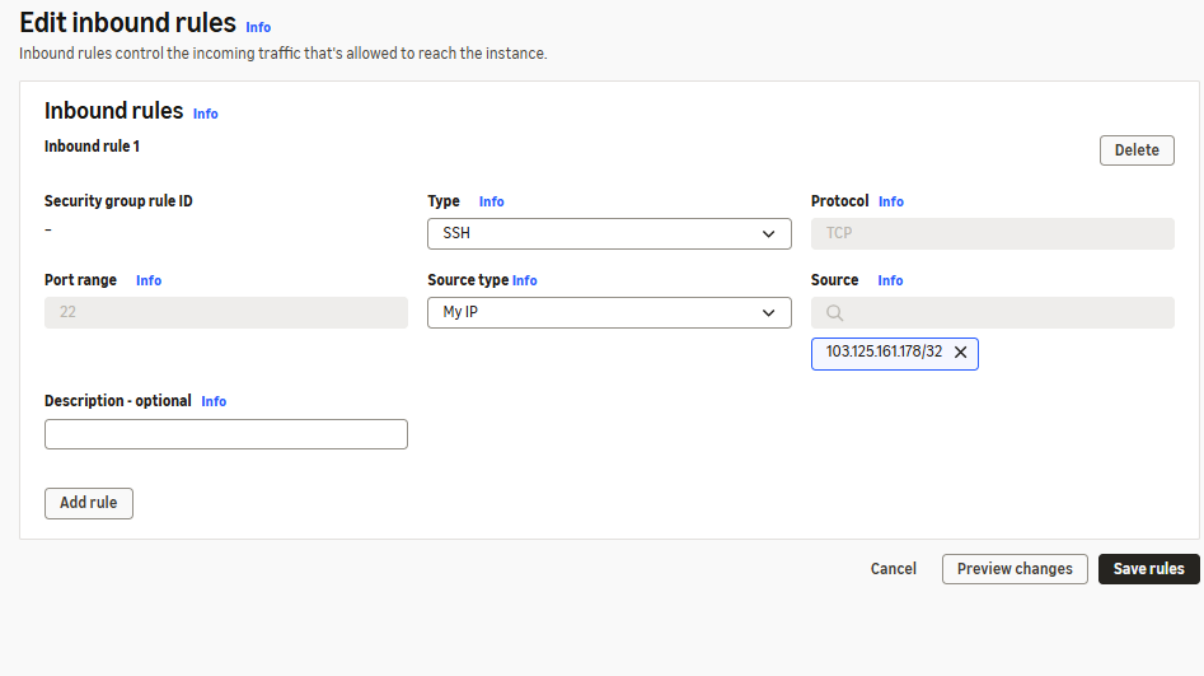
In SSH client

Description: *Selected the Connect option for the running My-First-EC2 instance to open the connection page and view the available methods for accessing the Linux instance.*



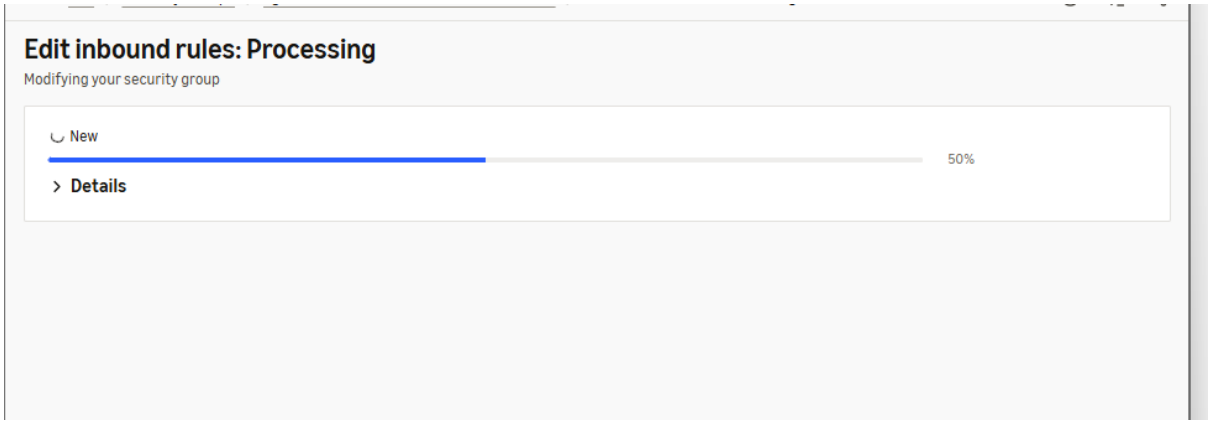
Selecting the Connection Method

Description: *The EC2 connection page displays different connection methods. Selected the EC2 Instance Connect option under the in-browser connection method and verified the public IP address and default username, ec2-user.*



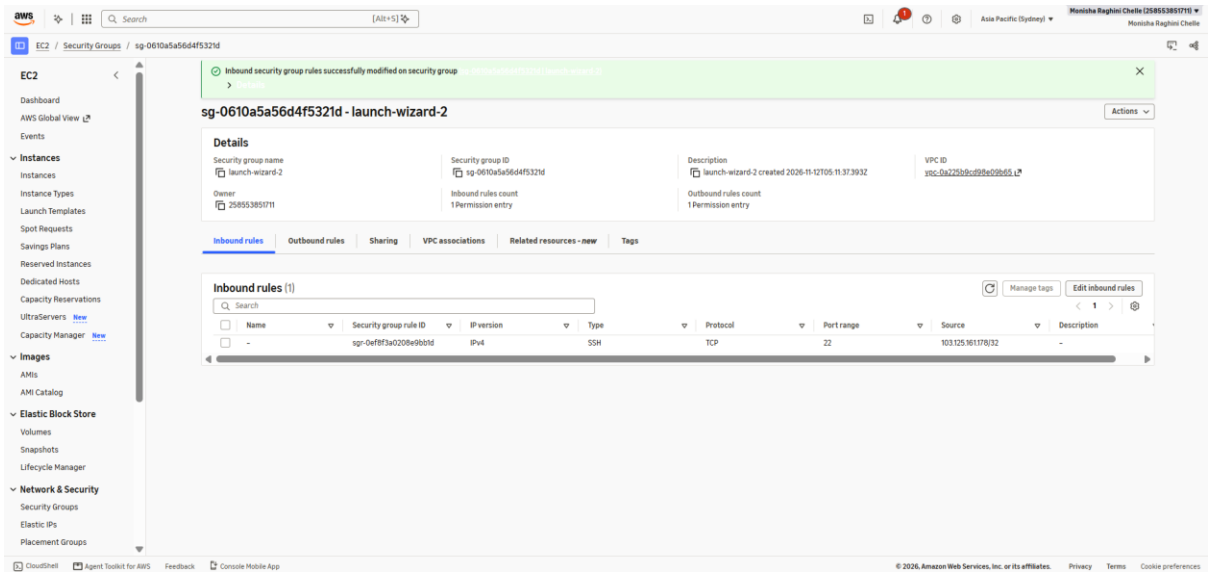
Configuring SSH Inbound Rules

Description: *Opened the security group's inbound rules and reviewed the SSH configuration. The rule uses TCP port 22 and allows access from My IP to enable secure SSH connections.*



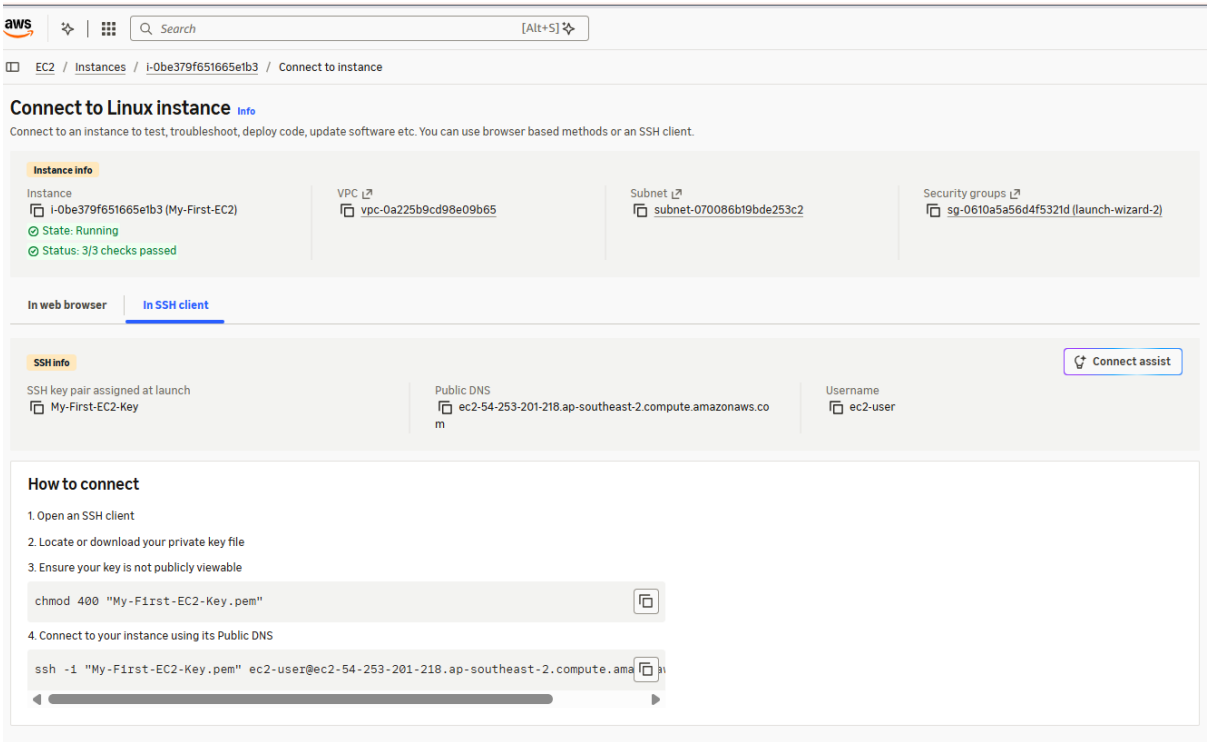
Saving the Security Group Rules

Description: *Saved the updated SSH inbound rule. AWS processed the changes to the security group to allow SSH access from the specified IP address.*



Verifying the SSH Rule

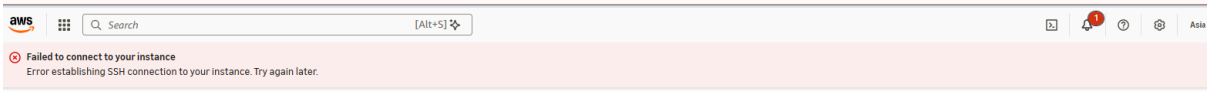
Description: *The security group page displayed the successfully modified inbound rule, confirming that SSH traffic is allowed through TCP port 22 from the configured IP address.*



Viewing SSH Connection Instructions

Description: *Selected the In SSH client tab to view the connection instructions. AWS displayed the private key filename, public DNS, username, and SSH command required to connect to the Linux instance.*

Here due to network

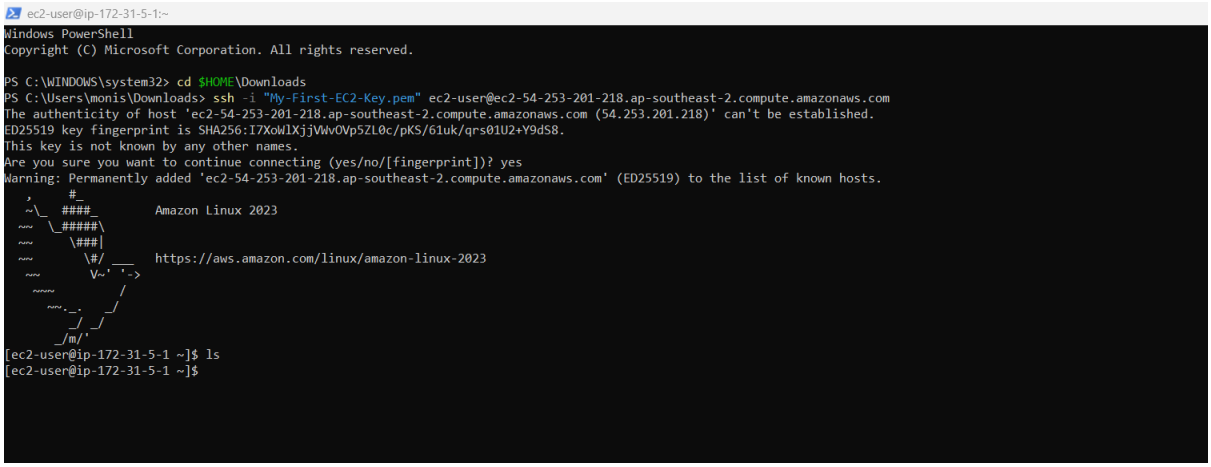


Failed to Connect to the EC2 Instance

Description: *After selecting Connect to establish an SSH connection to the running EC2 instance, the connection attempt failed. AWS displayed the error message, "Error establishing SSH connection to your instance. Try again later."*

Result: *The SSH connection was unsuccessful.*

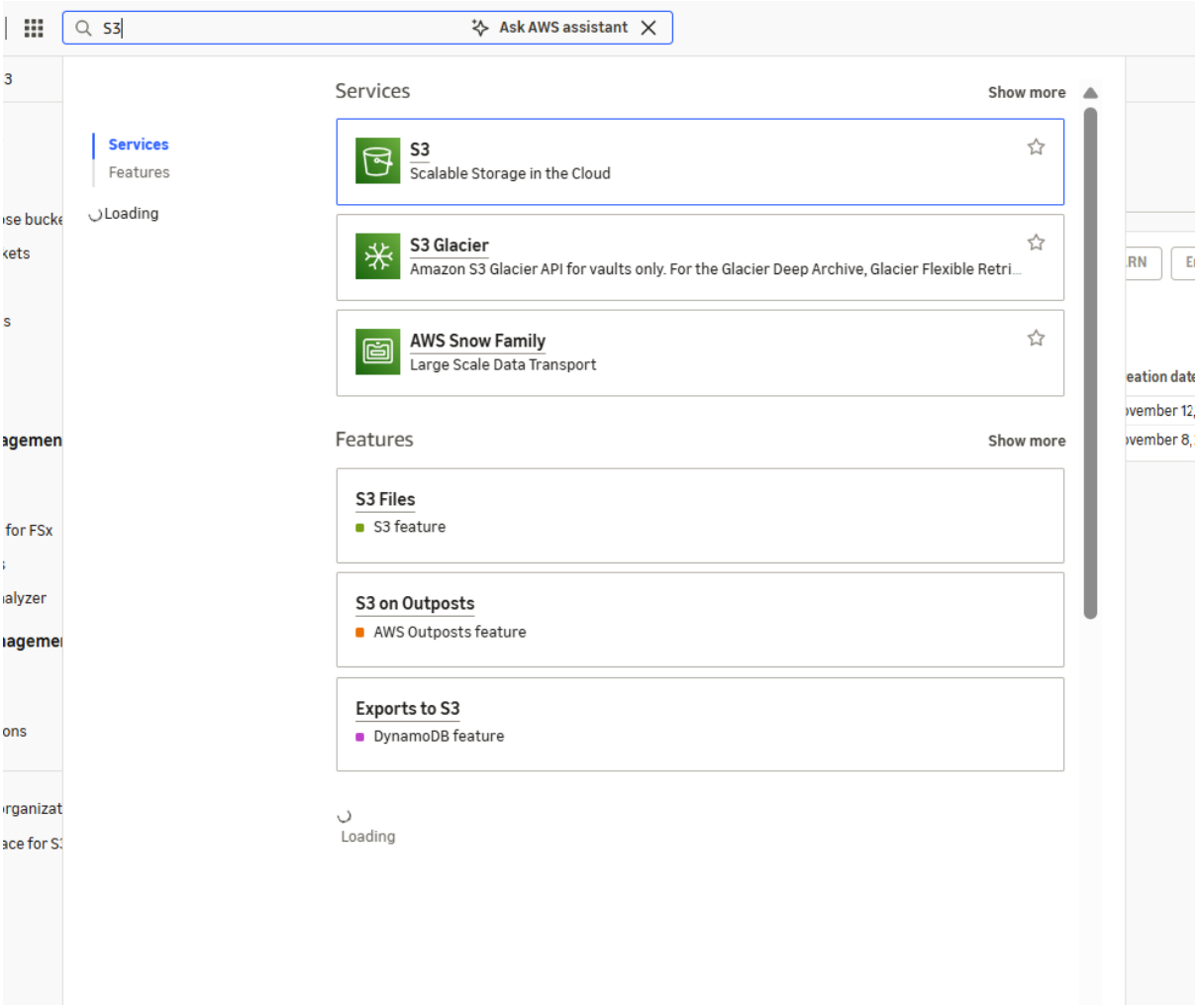
In Window Powershell



Windows PowerShell – Successful SSH Connection to EC2 Instance

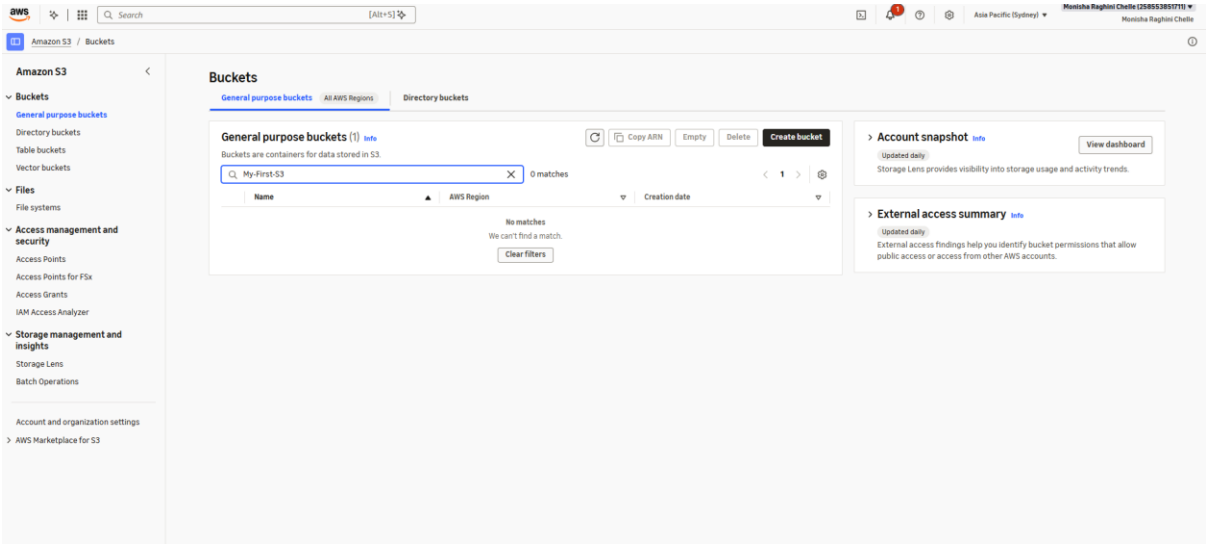
Description: *Opened Windows PowerShell and used the downloaded private key file (My-First-EC2-Key.pem) with the SSH command to connect to the EC2 instance. After accepting the host confirmation, the Amazon Linux 2023 welcome message appeared, and the EC2 command prompt was displayed successfully.*

2C. Creating an Amazon S3 Bucket and Uploading Objects



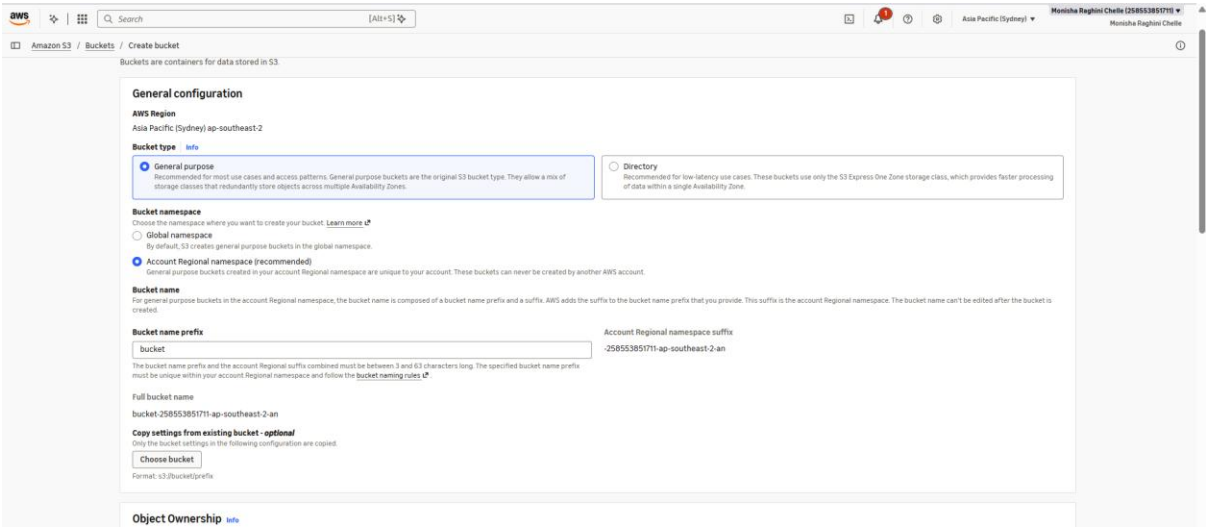
Searching for Amazon S3

Description: *Opened the AWS Management Console and searched for S3 in the search bar. Selected the Amazon S3 service to access cloud storage.*



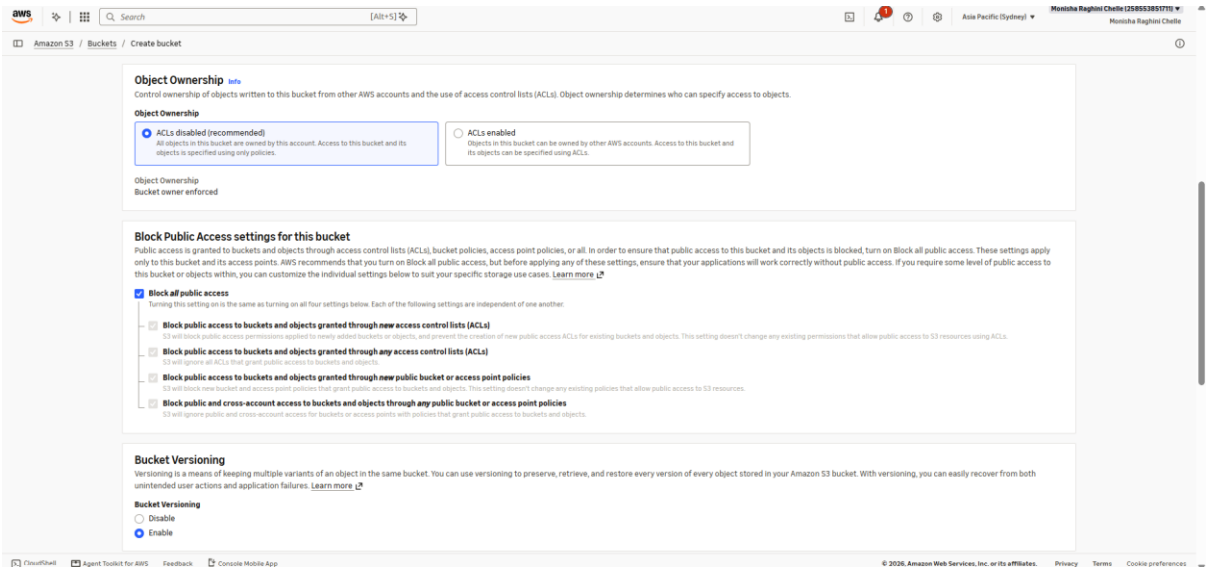
Amazon S3 – Buckets Page

Description: *Opened the Amazon S3 console and navigated to the Buckets section to view and manage S3 buckets.*



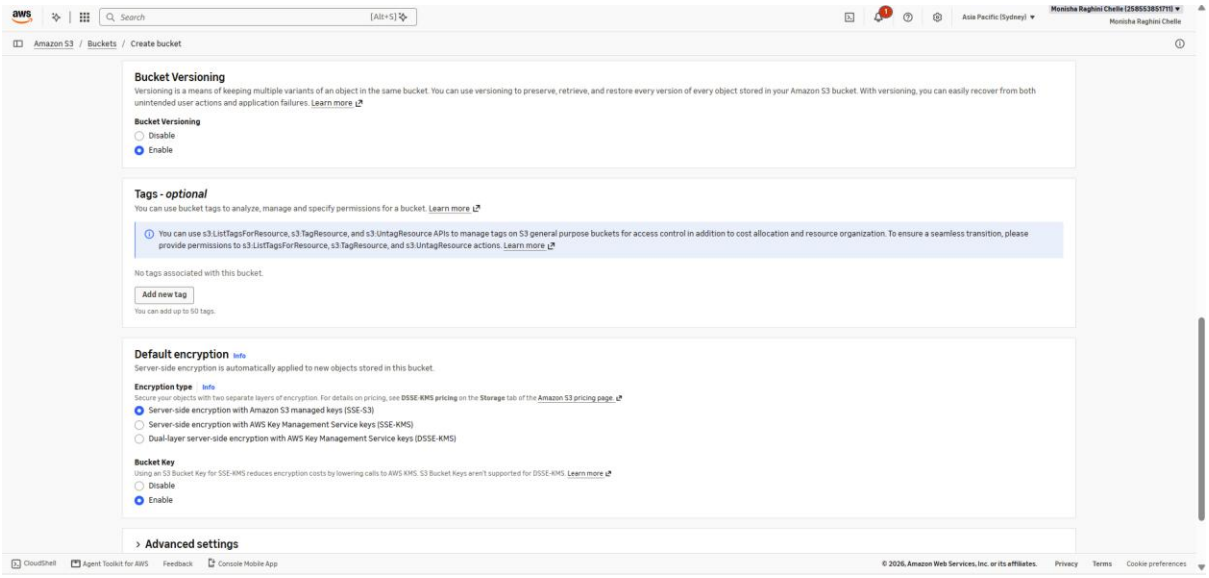
Create S3 Bucket – General Configuration

Description: *Selected Create bucket and configured the general settings by choosing the AWS Region and the general-purpose bucket type. Entered the required bucket name.*



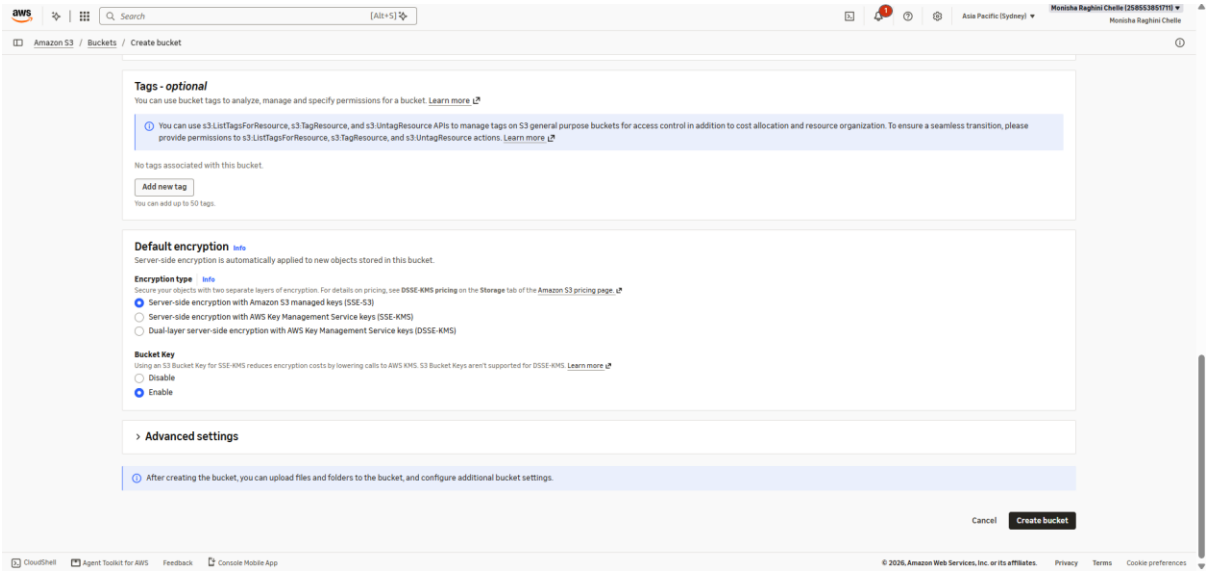
Object Ownership and Block Public Access

Description: *Configured the Object Ownership settings by keeping ACLs disabled and reviewed the Block Public Access settings. Kept Block Public Access enabled to prevent public access to the bucket.*



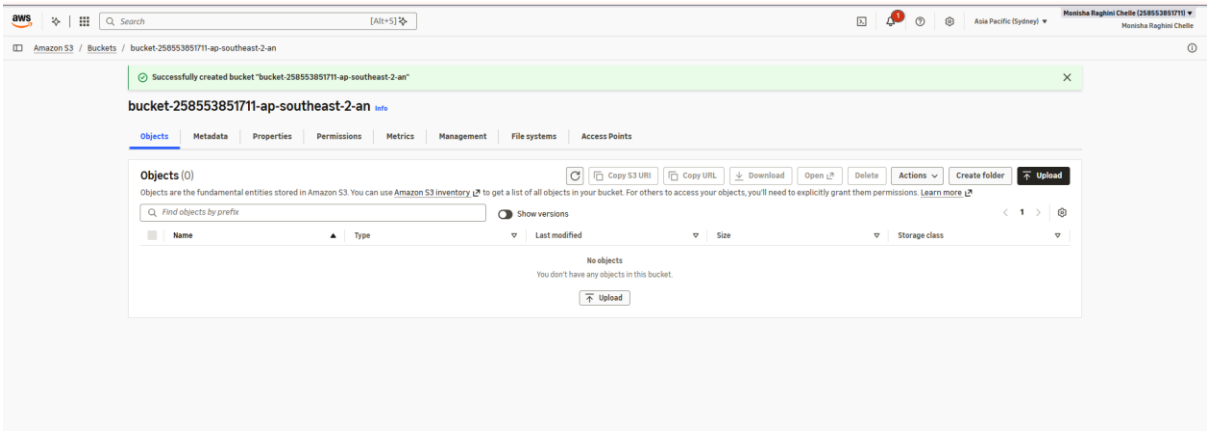
Bucket Versioning – Enabled

Description: *Enabled Bucket Versioning in the S3 bucket configuration. This feature helps preserve multiple versions of an object, allowing previous versions to be recovered if a file is accidentally modified or deleted.*



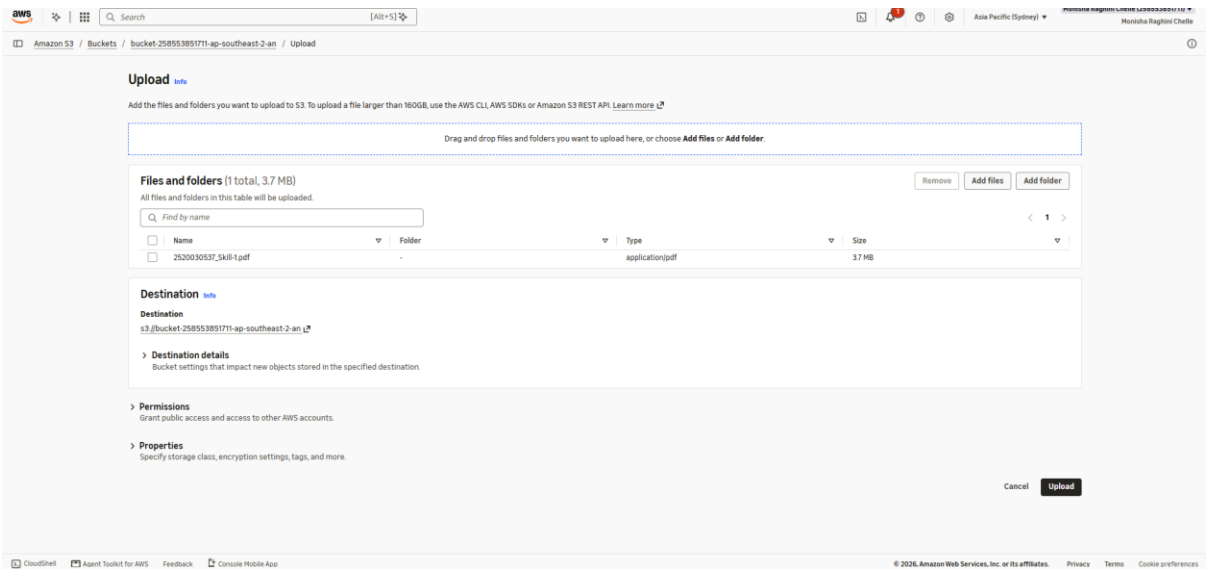
Tags and Default Encryption

Description: *Reviewed the optional Tags and Default Encryption settings. Kept the default server-side encryption option enabled to protect objects stored in the S3 bucket.*



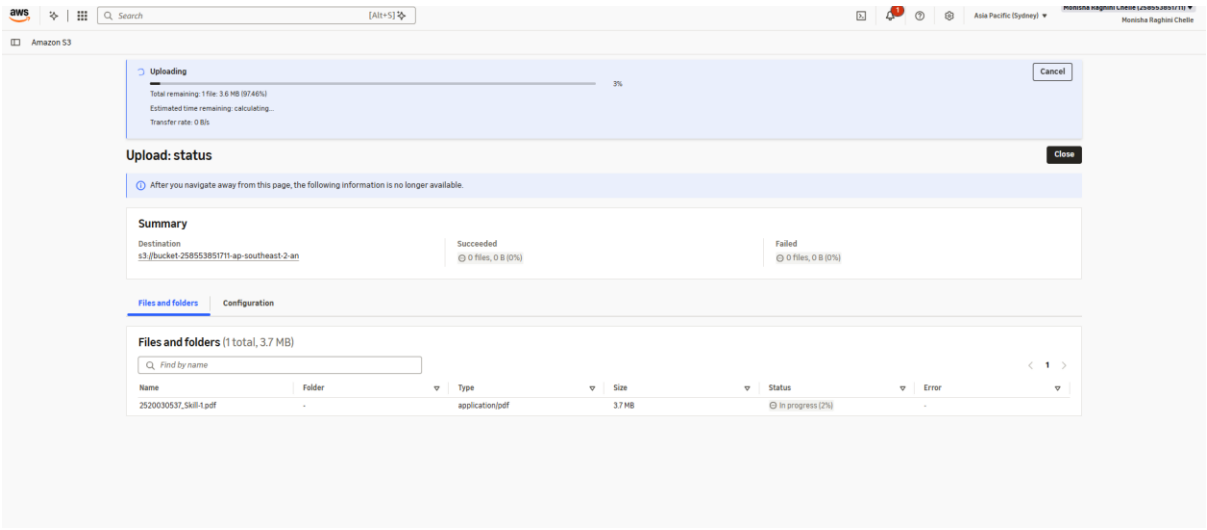
Successful Bucket Creation

Description: *Clicked Create bucket to complete the configuration. AWS displayed a success message confirming that the S3 bucket was created successfully.*



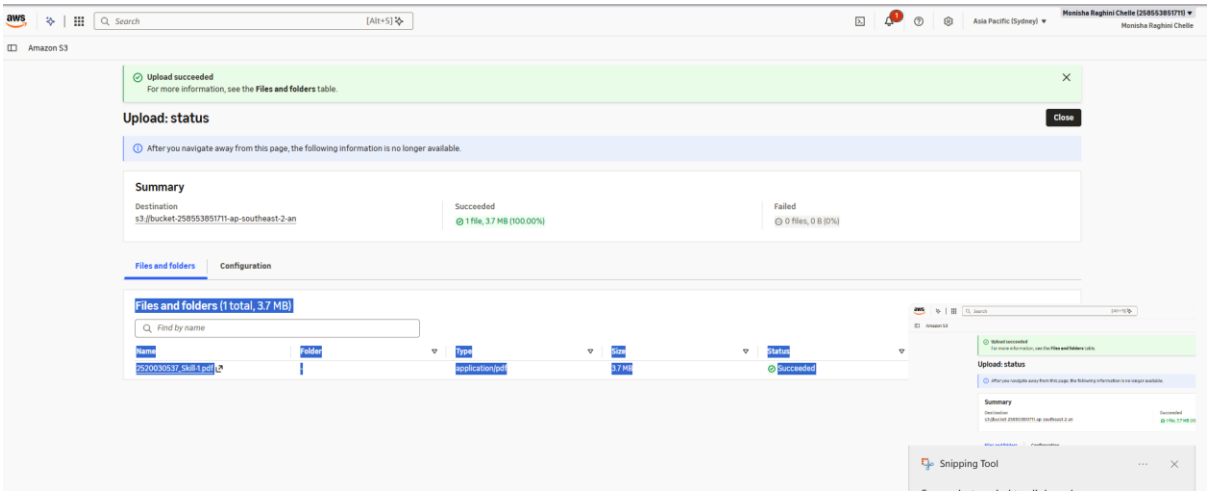
Selecting a File for Upload

Description: *Opened the newly created S3 bucket and selected Upload. Added the required PDF file from the computer and verified that it appeared in the upload list.*



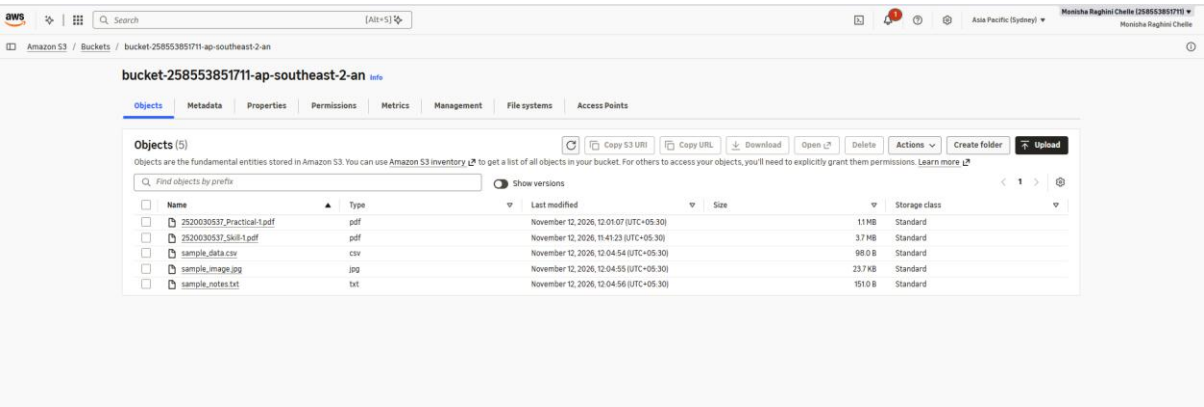
File Upload in Progress

Description: *Started the upload process by clicking Upload. AWS displayed the upload status and progress while transferring the selected PDF file to the S3 bucket.*



Successful File Upload

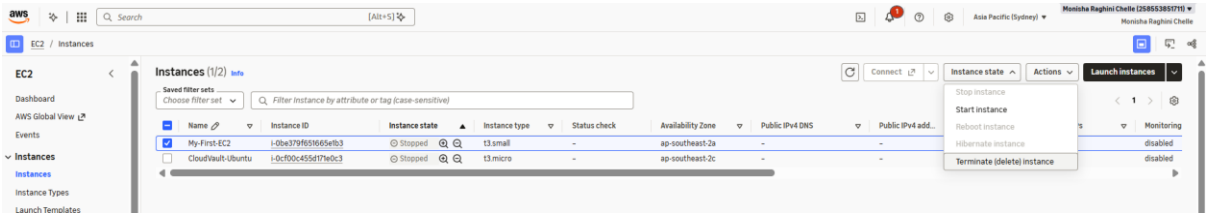
Description: *AWS displayed the Upload succeeded message, confirming that the PDF file was uploaded successfully to the S3 bucket. The uploaded file appeared in the Files and folders list.*



Amazon S3 – Verifying Uploaded Files

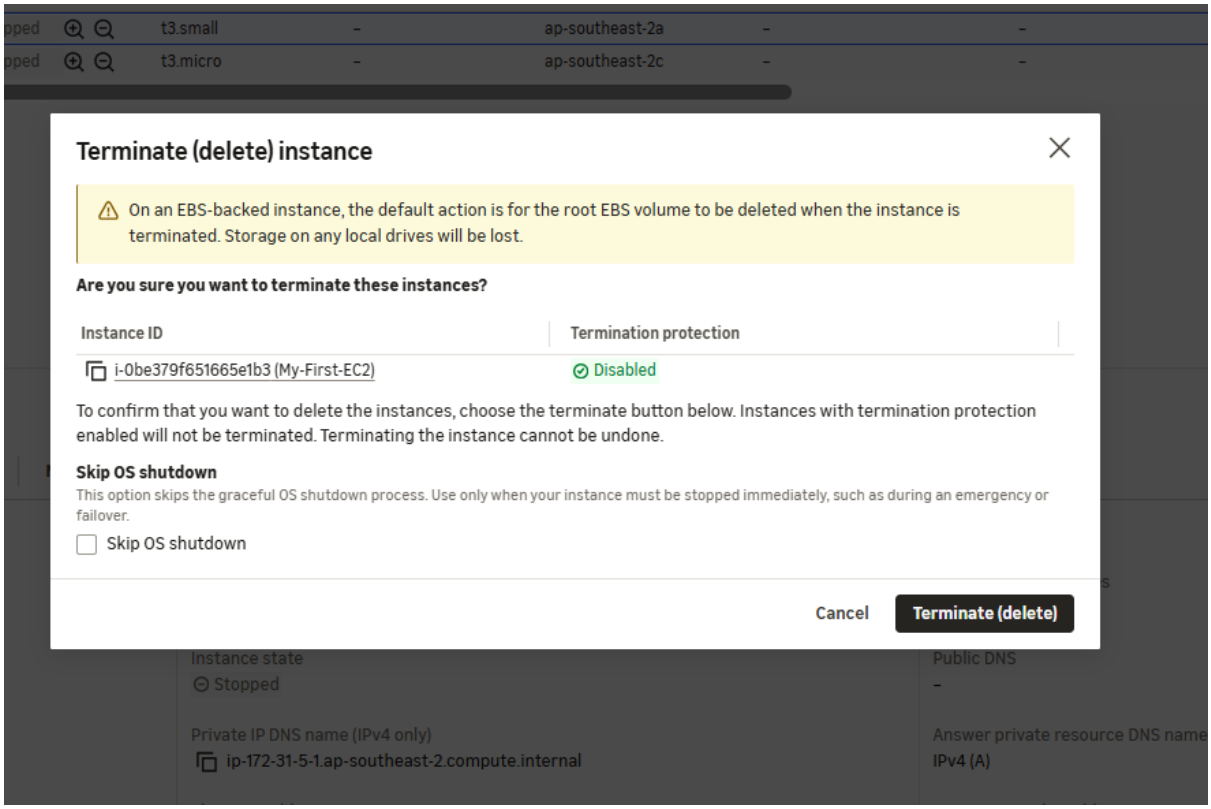
Description: *Opened the S3 bucket and checked the Objects section. Verified that all five uploaded files were displayed in the file list along with their file types, last modified dates, sizes, and storage class. This confirmed that the files were successfully stored in the S3 bucket.*

EC2 – Terminating the Instance



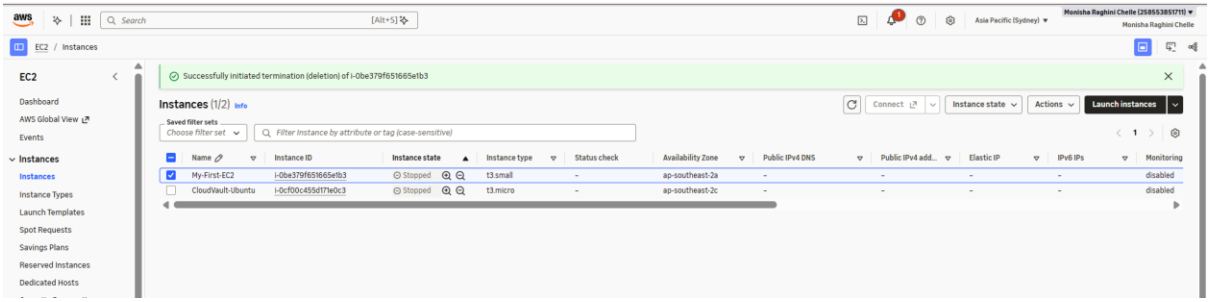
Selecting the EC2 Instance to Terminate

Description: *Opened the EC2 Instances page, selected the instance named My-First-EC2, and chose Terminate (delete) instance from the Instance state menu.*



Confirming Instance Termination

Description: *Reviewed the termination confirmation window and clicked Terminate (delete) to permanently delete the selected EC2 instance.*



Termination Request Submitted

Description: *AWS displayed a success notification confirming that the termination request was successfully initiated for the selected EC2 instance.*



Termination Status

Description: *Verified the EC2 Instances list, where the selected My-First-EC2 instance displayed the Terminated state, confirming that the instance was deleted successfully.*

Steps:

Part A: Enabling Multi-Factor Authentication (MFA)

1. Logged in to the AWS Management Console using the root account.
2. Opened the account security settings.
3. Navigated to the Multi-Factor Authentication (MFA) section.
4. Selected the option to assign an MFA device.
5. Chose an authenticator application as the MFA device.
6. Scanned the QR code using the authenticator application.

7. Entered the two consecutive verification codes.
8. Completed the setup and verified that MFA was enabled.

Part B: Creating the First Linux EC2 Instance

1. Logged in to the AWS Management Console.
2. Opened the EC2 service in the Asia Pacific (Sydney) region.
3. Clicked Launch instances to create a new virtual server.
4. Entered the instance name as My-First-EC2.
5. Selected the Amazon Linux 2023 AMI.
6. Selected the t3.small instance type.
7. Created a new key pair named My-First-EC2-Key and downloaded the .pem file.
8. Configured the network settings and enabled public IP assignment.
9. Created a security group and configured SSH access (port 22) from my IP address.
10. Reviewed the instance configuration and launched the instance.
11. Verified that the instance reached the Running state and that all three status checks passed.
12. Opened the Connect to instance page and selected the SSH client tab.
13. Used Windows PowerShell and the downloaded .pem key file to establish an SSH connection.
14. Stopped the EC2 instance after completing the practical.
15. Terminated the My-First-EC2 instance to avoid further instance-related charges.

Note: The browser-based SSH connection attempt initially failed. The SSH connection was subsequently established using Windows PowerShell.

Part C: S3 Bucket Creation and Uploading Objects

1. Opened the Amazon S3 service from the AWS Management Console.
2. Clicked Create bucket to create a new storage bucket.
3. Entered a globally unique bucket name.
4. Selected the AWS Region and reviewed the bucket settings.
5. Enabled bucket versioning and reviewed the default security and encryption settings.
6. Created the S3 bucket and opened it.
7. Clicked Upload to add files.
8. Selected five objects with different file extensions.
9. Uploaded the selected files to the bucket.
10. Verified that all five objects appeared in the bucket.

Output:

The following outputs were obtained during the practical:

1. Multi-Factor Authentication (MFA) was configured for the AWS account to provide an additional layer of security.
2. A Linux EC2 instance named My-First-EC2 was created using the Amazon Linux 2023 AMI and t3.small instance type.
3. A key pair named My-First-EC2-Key was created and downloaded in .pem format for SSH authentication.
4. The EC2 instance reached the Running state, and all three status checks passed.
5. An SSH connection was successfully established using Windows PowerShell and the downloaded private key.
6. The EC2 instance was stopped and subsequently terminated after completing the practical.
7. An S3 bucket was created for cloud-based file storage.
8. Bucket versioning was enabled, and five objects with different file extensions were uploaded successfully.
9. The practical demonstrated AWS account security, virtual server creation, SSH connectivity, and cloud object storage.